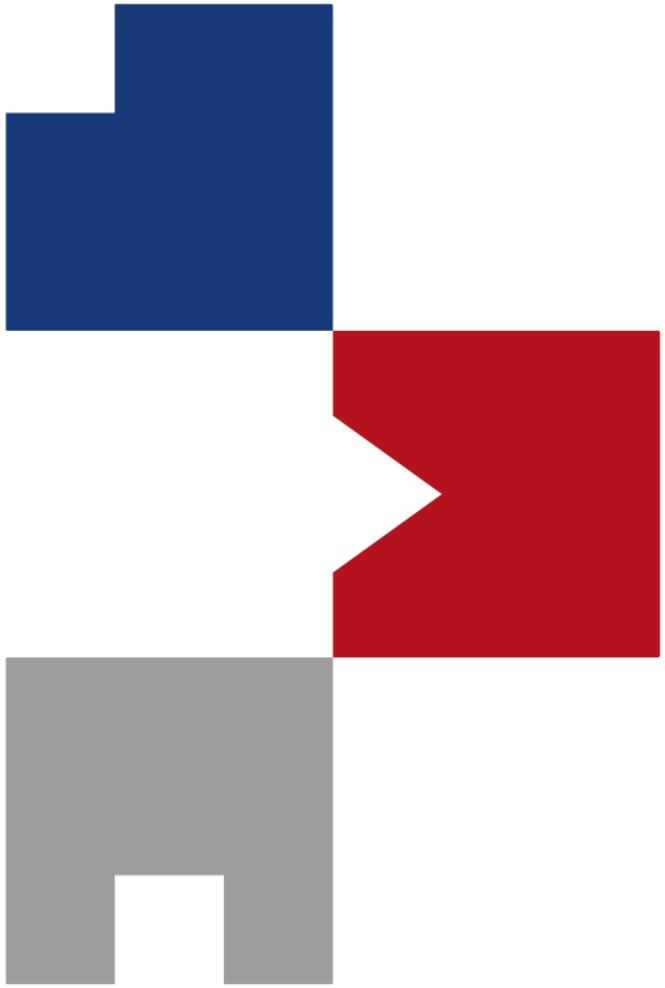
Note) Slides and example codes are available:
https://github.com/mint-lab/3dv_tutorial



An Invitation to 3D Vision:
Two-View Geometry

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sunglok@seoultech.ac.kr | https://mint-lab.github.io/

Review) Absolute Camera Pose Estimation

- Example) **Pose estimation (book) + camera calibration – initially given** K `[pose_estimation_book3.py]`

(due to wrong initial K)

- Load the cover image
- Extract ORB features
- Build **3D object points** $\mathbf{X}_1, \mathbf{X}_2, \ldots$

Pass tentative matches

- Grab an input image
- Extract **ORB features** $\mathbf{x}_1, \mathbf{x}_2, \ldots$
- Match them to the object features

[Outlier Rejection]
- Remove wrong matches using `cv::findHomography()`

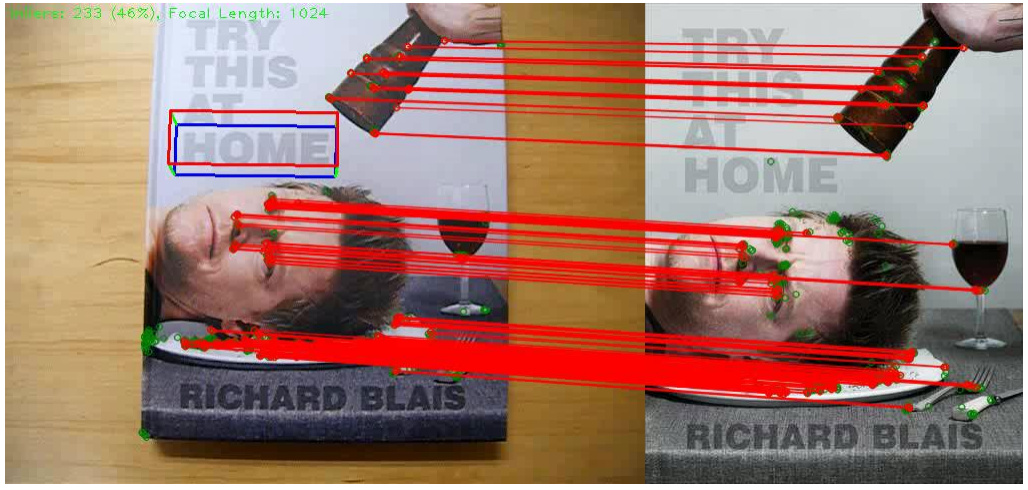
(free from K)

Pass correct matches

[Camera Pose Estimation]
- Estimate camera parameters using `cv::calibrateCamera()`

[Camera Pose Update]
- Intrinsic: K (initially known)
- Extrinsic: R, $\mathbf{t}$

Draw the box using `cv::projectPoints()`



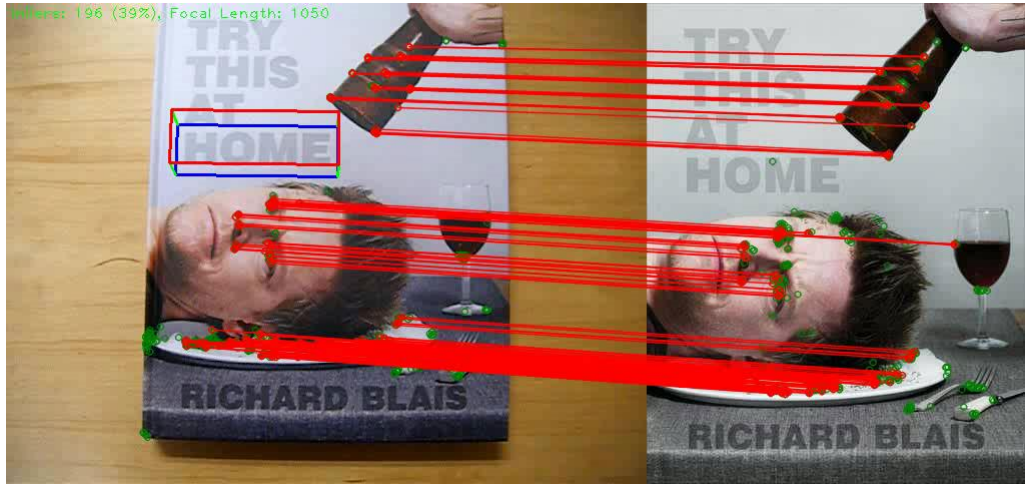
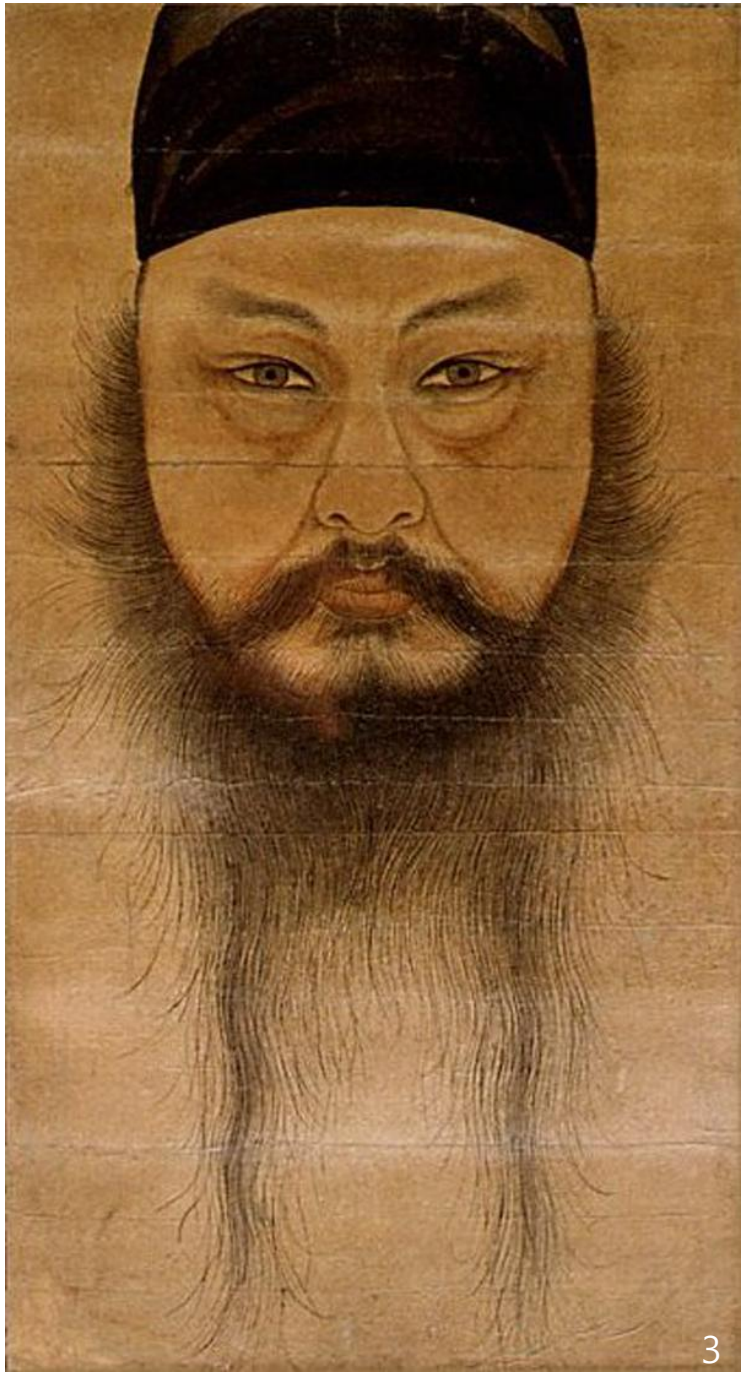


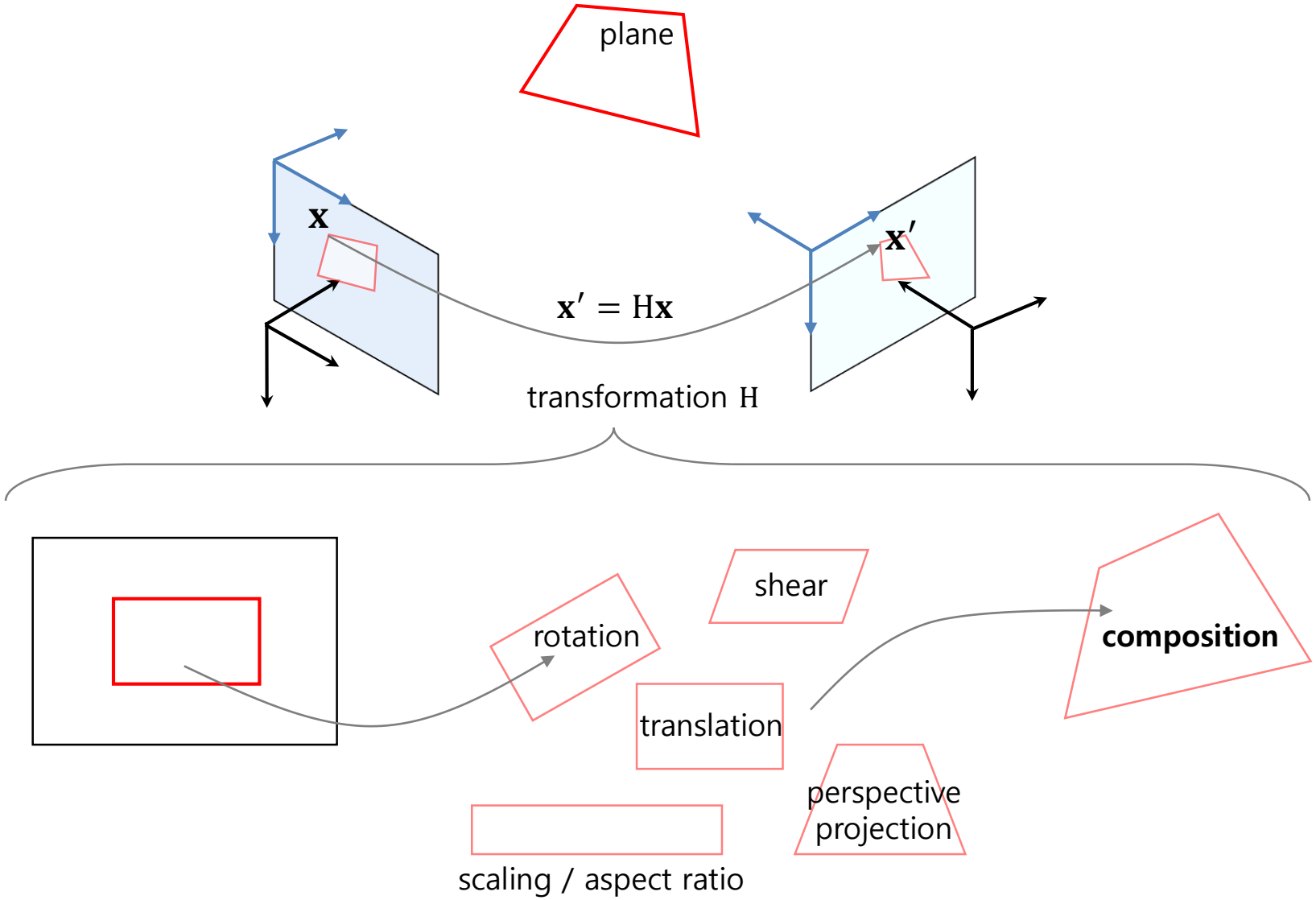
Table of Contents: Two-view Geometry

- **Planar Homography**
- **Epipolar Geometry**
  - Epipolar constraint
  - Fundamental and essential matrix
- **Relative Camera Pose Estimation**
- **Triangulation**

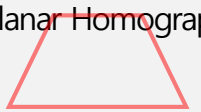


윤두서(1668-1715) 자화상, 국보 제240호
Korean National Treasure No. 240

Planar Homography



Planar Homography

		**Euclidean Transformation** (a.k.a. Rigid Transform) 	**Similarity Transformation** 	**Affine Transformation** 	**Projective Transformation** (a.k.a. Planar Homography) 
Matrix Forms H		$\begin{bmatrix}\cos\theta & -\sin\theta & t_x \\ \sin\theta & \cos\theta & t_y \\ 0 & 0 & 1\end{bmatrix}$	$\begin{bmatrix}s\cos\theta & -s\sin\theta & t_x \\ s\sin\theta & s\cos\theta & t_y \\ 0 & 0 & 1\end{bmatrix}$	$\begin{bmatrix}a_{11} & a_{12} & t_x \\ a_{21} & a_{22} & t_y \\ 0 & 0 & 1\end{bmatrix}$	$\begin{bmatrix}a_{11} & a_{12} & t_x \\ a_{21} & a_{22} & t_y \\ v_1 & v_2 & 1\end{bmatrix}$
DOF		3	4	6	8
Transformations - rotation - translation - scaling - aspect ratio - shear - perspective projection		 O O X X X X	 O O O X X X	 O O O O O X	 O O O O O O
Invariants - length - angle - ratio of lengths - parallelism - incidence - cross ratio		 O O O O O O	 X O O O O O	 X X X O O O	 X X X X O O

OpenCV APIs				`cv.getAffineTransform()`
				`cv.estimateRigidTransform()`
				-
				`cv.warpAffine()`
					`cv.getPerspectiveTransform()`
					-
					`cv.findHomography()`
					`cv.warpPerspective()`

Note) Similarly **3D transformations** (3D-3D geometry) are represented as **4x4 matrices**.

Note) Affine Transformation

- Examples

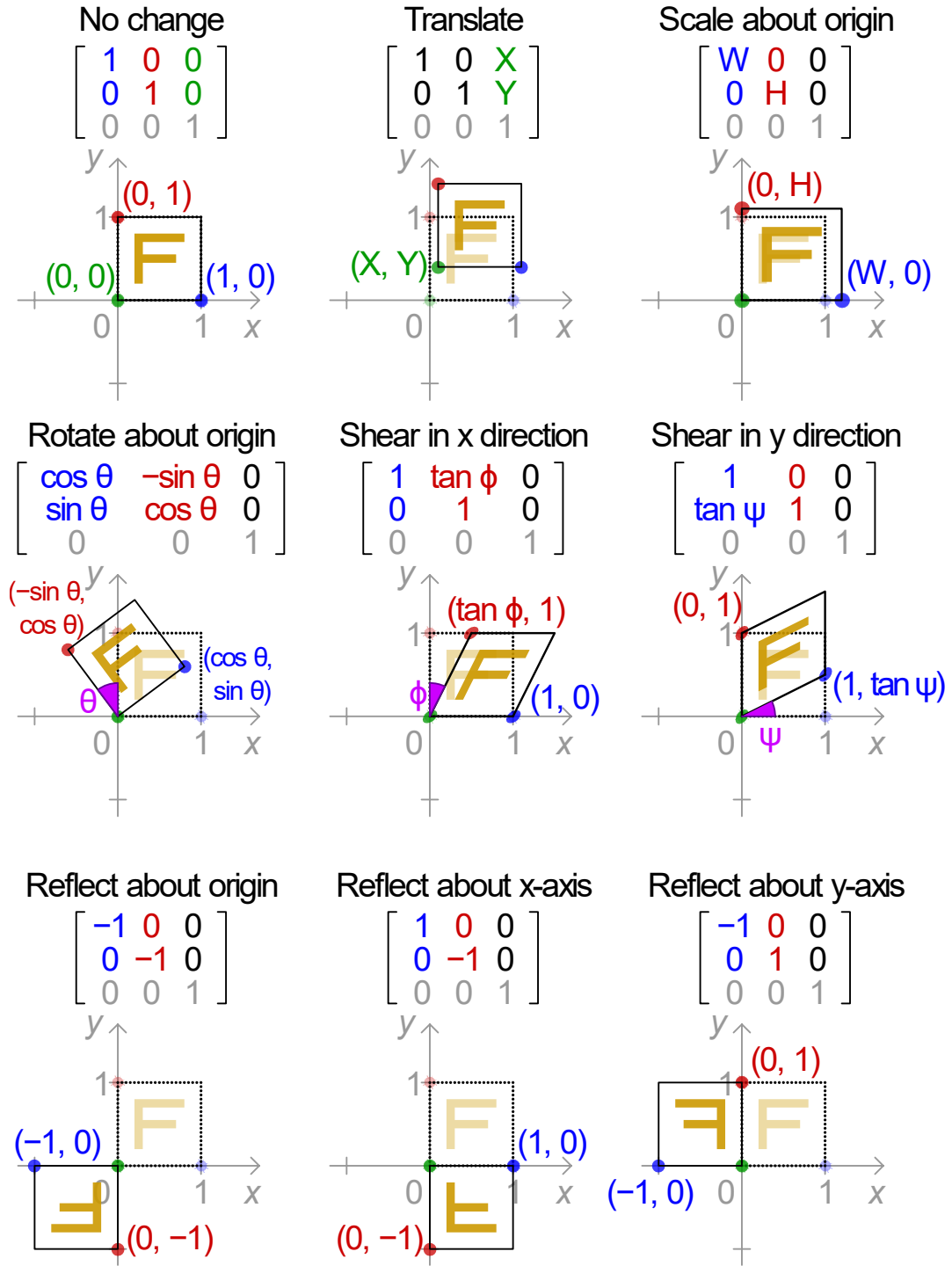
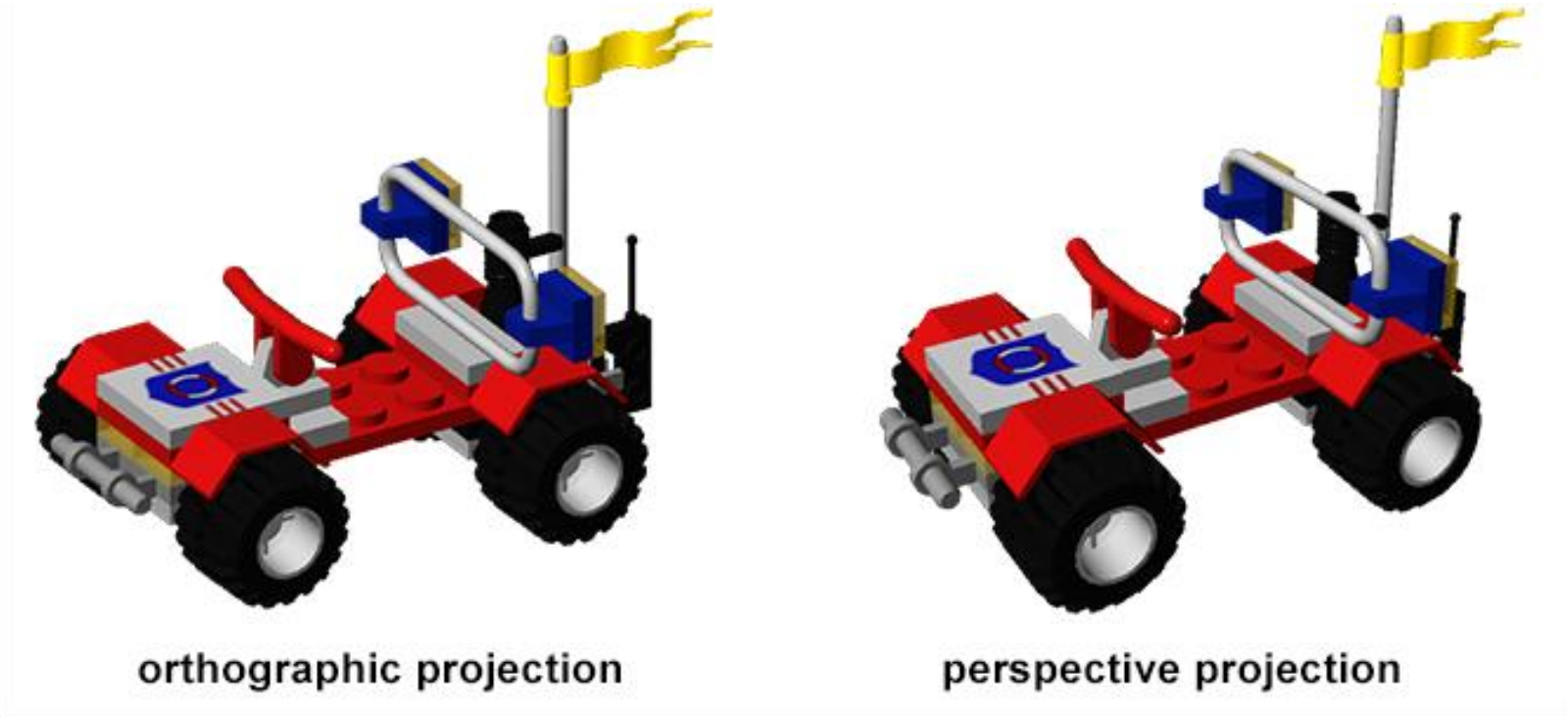


Image: Wikipedia

Note) Affine Transformation



- **Q) What is the difference between two images?**
  - Parallel projection vs. Perspective projection
    - Affine camera (a.k.a. weak perspective camera; $f = \infty$)
    - Less natural, but (sometimes) useful for technical visualization (e.g. CAD)

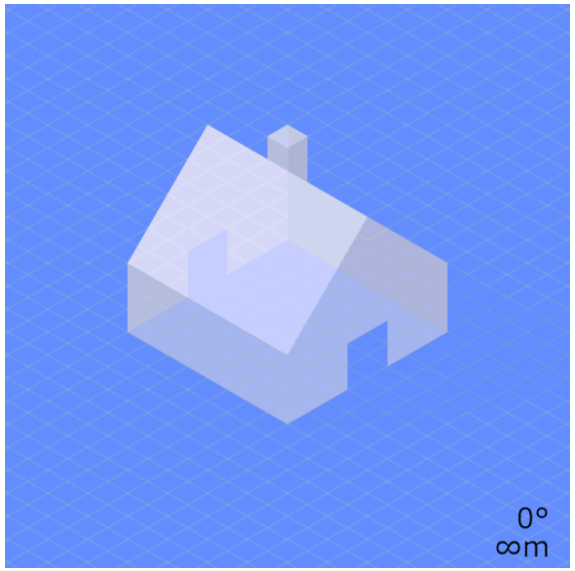
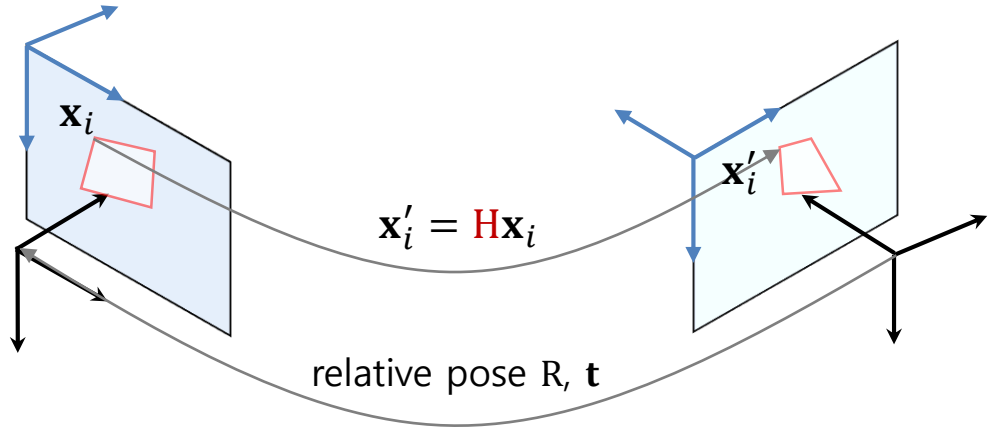


Image: Scatchpixel, Wikipedia

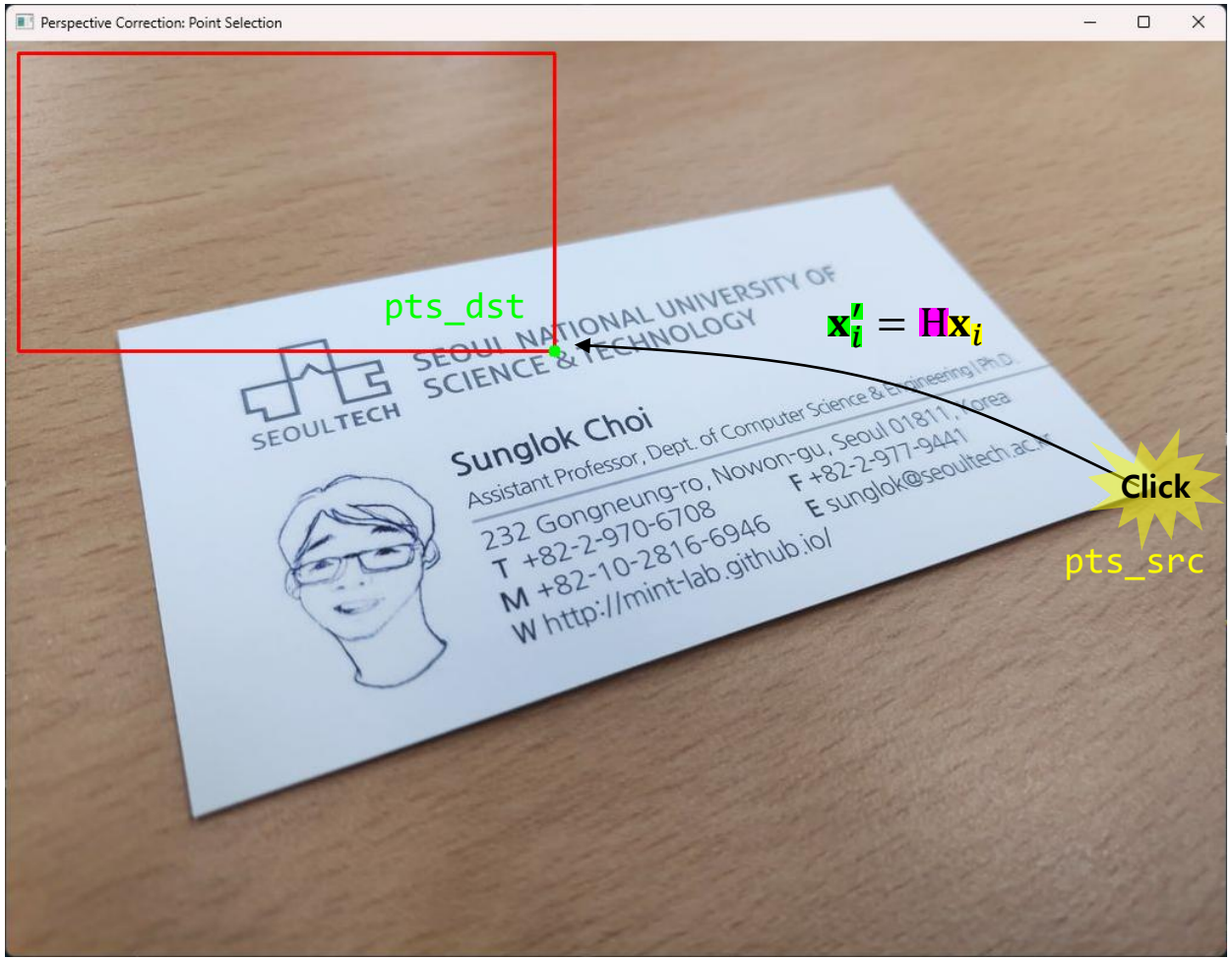
Planar Homography

- **Planar homography estimation**
  - Unknown: Planar homography H (8 DOF)
  - Given: Point correspondence $(\mathbf{x}_1, \mathbf{x}'_1)$, …, $(\mathbf{x}_n, \mathbf{x}'_n)$
  - Constraints: n x projective transformation $\mathbf{x}'_i = \mathrm{H}\mathbf{x}_i$
  - Solutions ($n \geq 4$) → 4-point algorithm
    - OpenCV: `cv.getPerspectiveTransform()` and `cv.findHomography()`
    - Note) Simpler transformations requires fewer correspondences.
      - Affine ($n \geq 3$), similarity ($n \geq 2$), Euclidean ($n \geq 2$)
  - Note) Planar homography can be decomposed as relative camera pose.
    - OpenCV: `cv.decomposeHomographyMat()`
    - The decomposition needs to know camera matrices.



Planar Homography

- Example) **Perspective distortion correction** `[perspective_correction.py]`





Planar Homography

- Example) **Perspective distortion correction** [perspective_correction.py]

```
def mouse_event_handler(event, x, y, flags, param):
    if event == cv.EVENT_LBUTTONDOWN:
        param.append((x, y))

if __name__ == '__main__':
    img_file = '../data/sunglok_card.jpg'
    card_size = (450, 250)
    offset = 10

    # Prepare the rectified points
    pts_dst = np.array([[0, 0], [card_size[0], 0], [0, card_size[1]], [card_size[0], card_size[1]]])

    # Load an image
    img = cv.imread(img_file)

    # Get the matched points from mouse clicks
    pts_src = []
    wnd_name = 'Perspective Correction: Point Selection'
    cv.namedWindow(wnd_name)
    cv.setMouseCallback(wnd_name, mouse_event_handler, pts_src)
    while len(pts_src) < len(pts_dst):
        img_display = img.copy()
        cv.rectangle(img_display, (offset, offset), (offset + card_size[0], offset + card_size[1]), (0, 0, 255), 2)
        idx = min(len(pts_src), len(pts_dst) - 1)
        cv.circle(img_display, offset + pts_dst[idx], 5, (0, 255, 0), -1)
        cv.imshow(wnd_name, img_display)
```

Planar Homography

- Example) **Perspective distortion correction** [perspective_correction.py]

```
if __name__ == '__main__':
    img_file = '../data/sunglok_card.jpg'
    card_size = (450, 250)
    offset = 10

    # Prepare the rectified points
    pts_dst = np.array([[0, 0], [card_size[0], 0], [0, card_size[1]], [card_size[0], card_size[1]]])

    # Load an image
    img = cv.imread(img_file)

    # Get the matched points from mouse clicks
    pts_src = []
    ...

    if len(pts_src) >= len(pts_dst):
        # Calculate planar homography and rectify perspective distortion
        H, _ = cv.findHomography(np.array(pts_src), pts_dst)
        img_rectify = cv.warpPerspective(img, H, card_size)

        # Show the rectified image
        cv.imshow('Perspective Distortion Correction: Rectified Image', img_rectify)
        cv.waitKey(0)

    cv.destroyAllWindows()
```

Planar Homography

- Example) **Planar image stitching** [image_stitching.py]

```
# Load two images
img1 = cv.imread('../data/hill01.jpg')
img2 = cv.imread('../data/hill02.jpg')

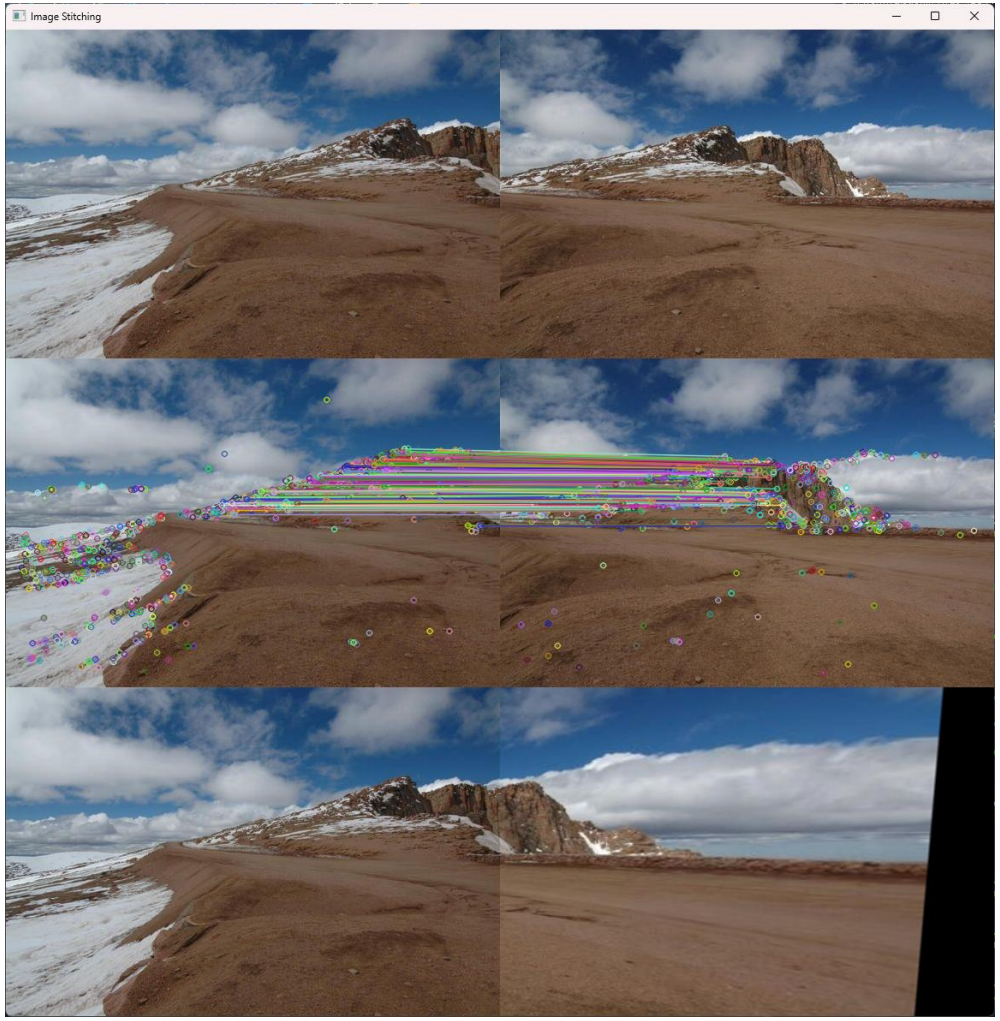
# Retrieve matching points
fdetector = cv.BRISK_create()
keypoints1, descriptors1 = fdetector.detectAndCompute(img1, None)
keypoints2, descriptors2 = fdetector.detectAndCompute(img2, None)

fmatcher = cv.DescriptorMatcher_create('BruteForce-Hamming')
match = fmatcher.match(descriptors1, descriptors2)

# Calculate planar homography and merge two images
pts1, pts2 = [], []
for i in range(len(match)):
    pts1.append(keypoints1[match[i].queryIdx].pt)
    pts2.append(keypoints2[match[i].trainIdx].pt)
pts1 = np.array(pts1, dtype=np.float32)
pts2 = np.array(pts2, dtype=np.float32)

H, inlier_mask = cv.findHomography(pts2, pts1, cv.RANSAC)
img_merged = cv.warpPerspective(img2, H, (img1.shape[1]*2, img1.shape[0]))
img_merged[:,:img1.shape[1]] = img1 # Copy

# Show the merged image
img_matched = cv.drawMatches(img1, keypoints1, img2, keypoints2, match, None, None, None,
```



Planar Homography

- Example) **Planar image stitching → *Gigapixel* images** (with 10^9+ pixels)



Image: Wikimedia Commons

Planar Homography

- Example) ~~Planar~~ **image stitching → *Panoramic* images** (with wide-angle view)



Image: OpenPano

Image: Seada Haskić, John David Lynn Brosz, Wikipedia

Beyond Planar Homography

- **Projection models for *panoramic* images** (with wide-angle view)



Planar/Perspective view
z = const



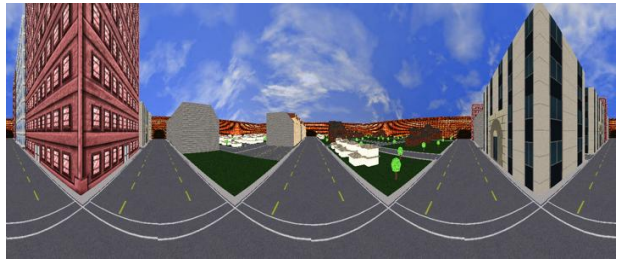
+ Natural (preserve straight lines)
- Limited field-of-view (max <180°)



3D Model



Cylindrical view
$x^2 + z^2 = const$



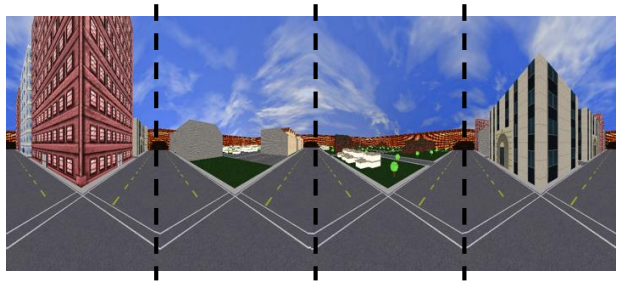


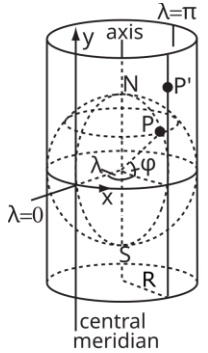
Spherical view
$x^2 + y^2 + z^2 = const$

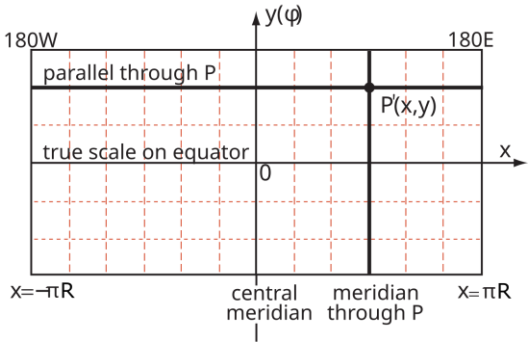




Cubic view
$-const \le x \le +const$
$-const \le y \le +const$
$z = \pm const$ ...







How do we represent a sphere in 2D?
e.g. Mercator projection

Planar Homography

- Example) **2D video stabilization** [video_stabilization.py]

```
# Open a video and get the reference image and feature points
video = cv.VideoCapture('../data/traffic.avi')

_, gray_ref = video.read()
if gray_ref.ndim >= 3:
    gray_ref = cv.cvtColor(gray_ref, cv.COLOR_BGR2GRAY)
pts_ref = cv.goodFeaturesToTrack(gray_ref, 2000, 0.01, 1

# Run video stabilization
while True:
    # Read an image from `video`
    ...

    # Extract optical flow and calculate planar homography
    pts, status, error = cv.calcOpticalFlowPyrLK(gray_ref, gray, pts_ref, None)
    tracked_mask = status.ravel().astype(bool)
    pts, ref = pts[tracked_mask], pts_ref[tracked_mask]
    H, inlier_mask = cv.findHomography(pts, ref, cv.RANSAC)

    # Synthesize a stabilized image
    warp = cv.warpPerspective(img, H, (img.shape[1], img.shape[0]))

    # Show the original and stabilized images together
    for pt, pt_ref, inlier in zip(pts, ref, inlier_mask):
        color = (0, 0, 255) if inlier else (0, 127, 0)
        cv.line(img, pt.flatten().astype(np.int32),
```

A shaky CCTV video



(Red: Inlier, Green: Outlier)

Planar Homography

- Assumption) **The same plane is observed from two viewpoints.**
  - Perspective distortion correction: A complete plane









Planar Homography

- Assumption) **A <u>plane</u> is observed by two views.**
  - Perspective distortion correction: A complete plane
  - Planar image stitching: An approximated plane (← distance ≫ depth variation)
  - 2D video stabilization: An approximated plane (← small motion)









Epipolar Geometry

- **Epipolar constraint**

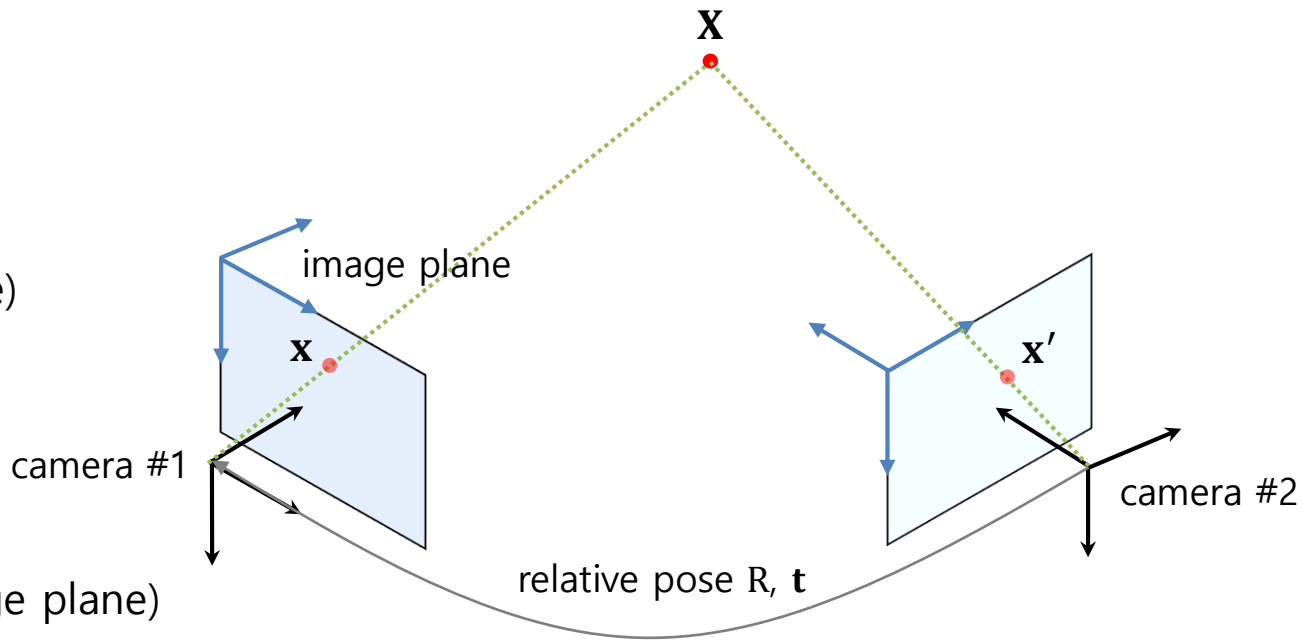
$\mathbf{x}'^{\mathsf{T}}\mathrm{F}\mathbf{x} = 0$ (F: Fundamental matrix on the image plane)

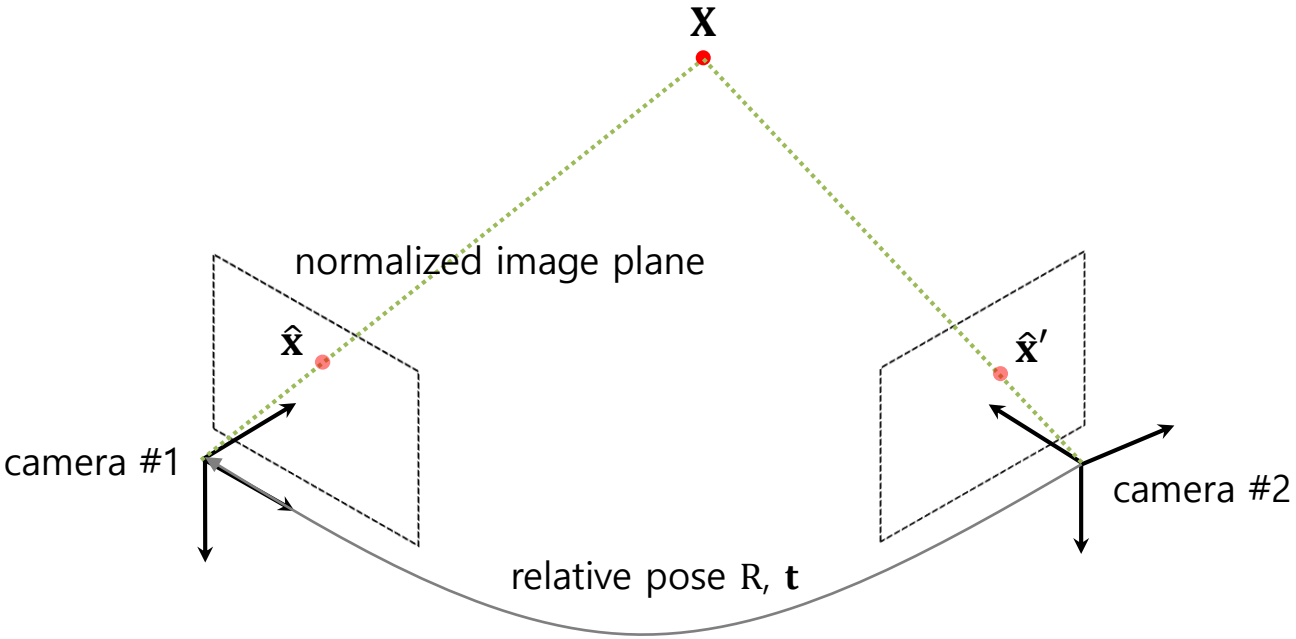
$\updownarrow \quad (\mathbf{x} = \mathrm{K}\hat{\mathbf{x}})$

$\hat{\mathbf{x}}'^{\mathsf{T}}\mathrm{K}'^{\mathsf{T}}\mathrm{F}\mathrm{K}\hat{\mathbf{x}} = 0$

$\updownarrow \quad (\mathrm{E} = \mathrm{K}'^{\mathsf{T}}\mathrm{F}\mathrm{K})$

$\hat{\mathbf{x}}'^{\mathsf{T}}\mathrm{E}\hat{\mathbf{x}} = 0$ (E: Essential matrix on the normalized image plane)





Epipolar Geometry

- **Epipolar constraint: Derivation**

$$\mathbf{X}' = \mathrm{R}\mathbf{X} + \mathbf{t}$$

$$\downarrow \quad (\mathbf{X} = \lambda\hat{\mathbf{x}})$$

$$\lambda'\hat{\mathbf{x}}' = \lambda\mathrm{R}\hat{\mathbf{x}} + \mathbf{t}$$

$$\mathbf{t} \times \downarrow$$

$$\lambda'\mathbf{t} \times \hat{\mathbf{x}}' = \lambda\mathbf{t} \times \mathrm{R}\hat{\mathbf{x}}$$

$$\hat{\mathbf{x}}' \cdot \downarrow$$

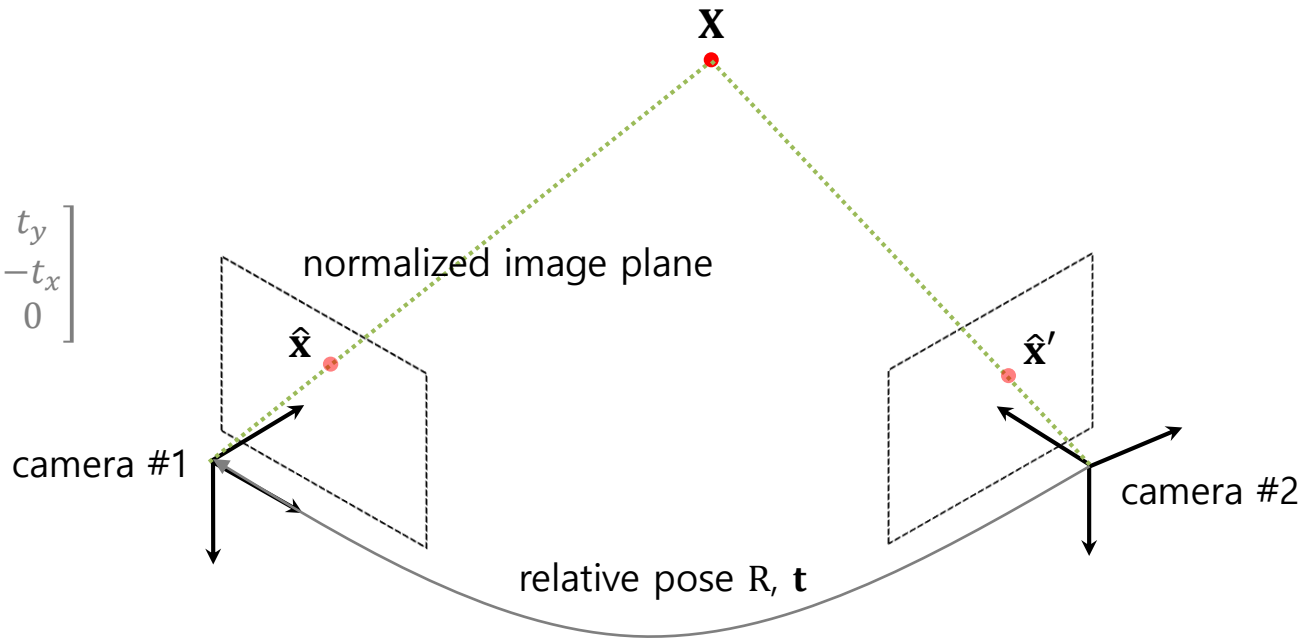
$$\lambda'\hat{\mathbf{x}}' \cdot (\mathbf{t} \times \hat{\mathbf{x}}') = \lambda\hat{\mathbf{x}}' \cdot \mathbf{t} \times \mathrm{R}\hat{\mathbf{x}}$$

$$\downarrow \quad (\hat{\mathbf{x}}' \cdot (\mathbf{t} \times \hat{\mathbf{x}}') = 0)$$

$$\lambda\hat{\mathbf{x}}' \cdot \mathbf{t} \times \mathrm{R}\hat{\mathbf{x}} = 0$$

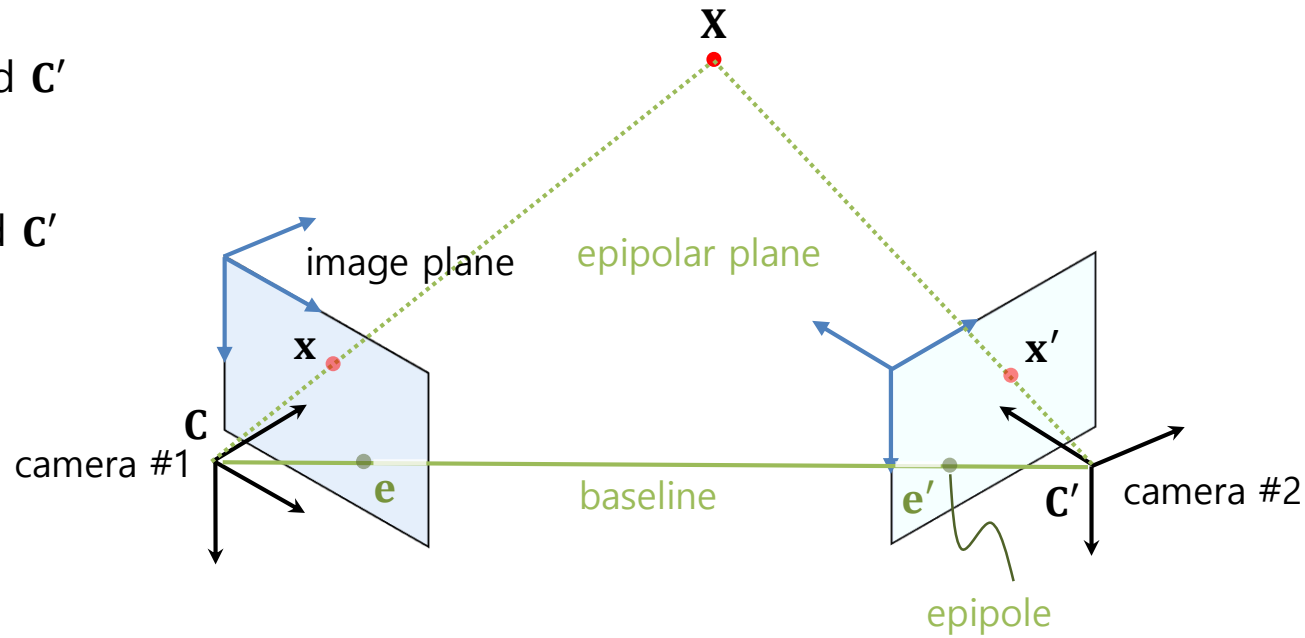
$$\downarrow \quad (\mathrm{E} = \mathbf{t} \times \mathrm{R} \text{ or } \mathrm{E} = [\mathbf{t}]_\times \mathrm{R}) \quad \text{Note) } [\mathbf{t}]_\times = \begin{bmatrix} 0 & -t_z & t_y \\ t_z & 0 & -t_x \\ -t_y & t_x & 0 \end{bmatrix}$$

$$\hat{\mathbf{x}}'\mathrm{E}\hat{\mathbf{x}} = 0$$



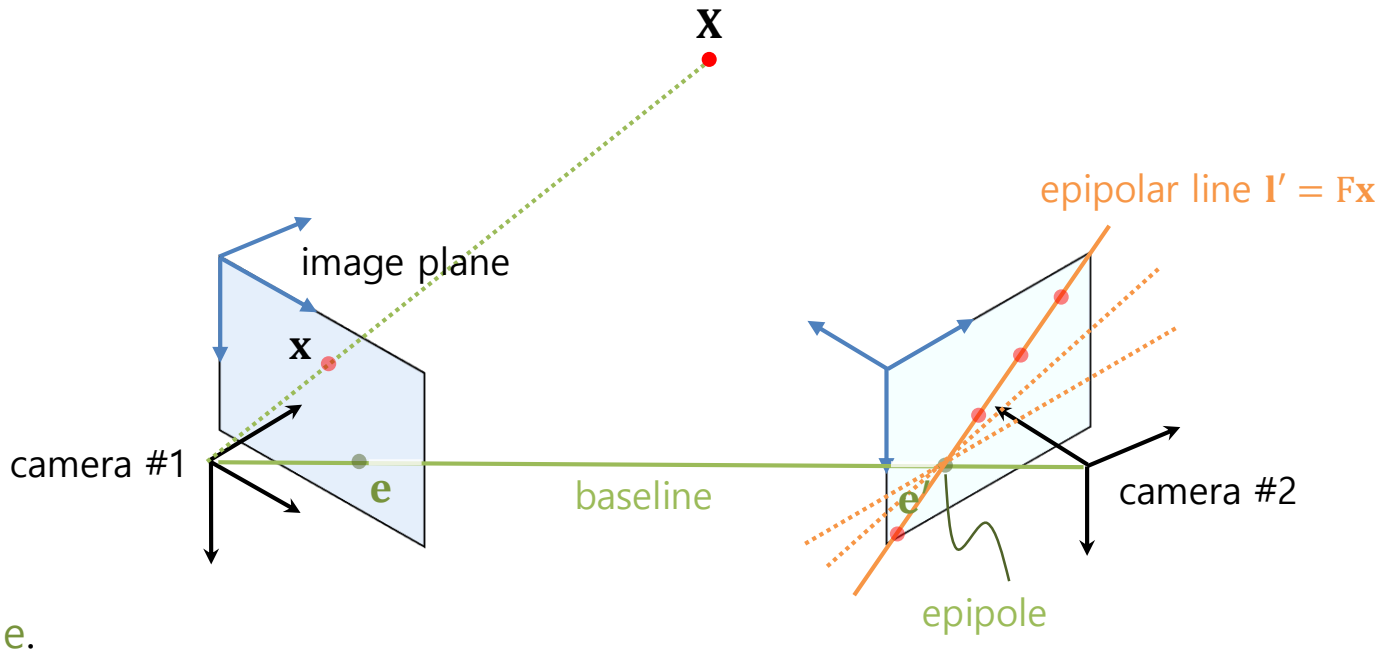
Epipolar Geometry

- **Epipolar geometry**
  - Baseline
    - Distance between two camera centers, $\mathbf{C}$ and $\mathbf{C}'$
  - Epipolar plane
    - Plane generated from three points, $\mathbf{X}$, $\mathbf{C}$, and $\mathbf{C}'$
  - Epipole (a.k.a. epipolar point)
    - Projection of other camera centers
    - $\mathbf{e} = \mathrm{P}\mathbf{C}'$ and $\mathbf{e}' = \mathrm{P}'\mathbf{C}$
      - e.g. $\mathrm{P} = \mathrm{K}[\mathrm{I}|0]$ and $\mathrm{P}' = \mathrm{K}'[\mathrm{R}|\mathbf{t}]$
        - $\mathbf{e} = -\mathrm{K}\mathrm{R}^{\mathrm{T}}\mathbf{t}$ and $\mathbf{e}' = \mathrm{K}'\mathbf{t}$

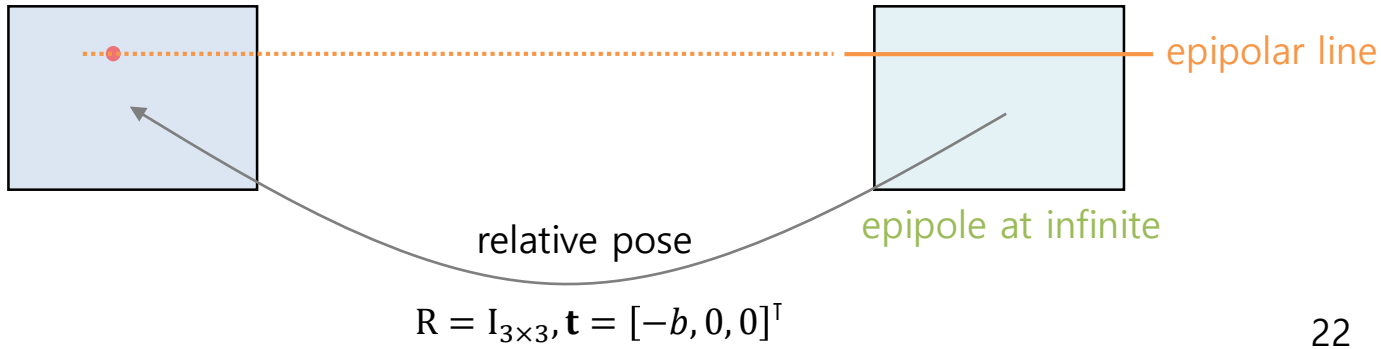


Epipolar Geometry

- **Epipolar geometry**
  - Epipolar line
    - $\mathbf{x}'^{\mathsf{T}}\mathrm{F}\mathbf{x} = 0 \rightarrow \mathbf{x}'^{\mathsf{T}}\mathbf{l}' = 0$ where $\mathbf{l}' = \mathrm{F}\mathbf{x}$
      - $\mathbf{x}'$ will lie on the line $\mathbf{l}'$.
    - $\mathbf{x}'^{\mathsf{T}}\mathrm{F}\mathbf{x} = 0 \rightarrow \mathbf{l}^{\mathsf{T}}\mathbf{x} = 0$ where $\mathbf{l} = \mathrm{F}^{\mathsf{T}}\mathbf{x}'$
      - $\mathbf{x}$ will lie on the line $\mathbf{l}$.
    - The search space for feature matching is reduced from a 2D image plane to the 1D epipolar line.
  - Note) Every epipolar line intersects at the epipole.
    - $\mathrm{F}\mathbf{e} = 0$
    - $\mathbf{e}$ is the null space of F.



Special case) Stereo cameras



Epipolar Geometry

- Fun) The Fundamental Matrix Song

The Fundamental Matrix Song

If you like this, you'll probably also be interested in the RANSAC song and We are the NeRFs.

The fundamental matrix
Used in stereo geometry
A matrix with nine entries
It's square with size 3 by 3
Has seven degrees of freedom
It has a rank deficiency
It's only of rank two
Call the matrix F and you'll see...

Two points that correspond
Column vectors called x and x-prime
x-prime transpose times F times x
Equals zero every time

The epipolar constraint
Involves epipolar lines
Postmultiplying F by x
Results in vector l-prime
It's the epipolar line
In the other view passing through x-prime
A three component vector
Of homogeneous design

The left and right nullspaces of F
Are the epipoles e-prime and e
All of the epipolar lines
Should pass through these

Here's a linear estimation example:
Take a set of 8 point samples
Construct a matrix, take the SVD
And the elements of F are in the last column of V

If you try to estimate
F with a coplanar set of points
Your sample set will be degenerate
And will not bring you joy

When doing the estimation
If you don't perform rank deprivation
Your epipolar lines
And the epipoles will not coincide

But if your scene has three views
The trifocal tensor is what you'd use
Constraints from the third view act like glue
That can't be determined from just two views



You may also be interested in 2.71828183: The e Song.
Video with Japanese subtitles
Download a high-quality version of the video (about 25MB) (requires Xvid codec)
Download video with English subtitles
Download karaoke video (vocals removed)
Download the MP3 (about 4MB)
Download the Opera House sequence

Thanks to:

Dr Peter Kovesi, whose code turned Matlab into a video effects tool,
Peter Pakulski, the AviSynth guru,
ilesoft, whose Sydney Harbour Bridge model is scarily accurate, and
Payashim, for the Japanese translation.

Related sites

- Hartley and Zisserman's "Bible": Multiple View Geometry in Computer Vision
- Peter Kovesi's Matlab functions for Computer Vision

Email: fmatrix at danielwedge dot com
Feel free to play this in lectures etc, you have my permission (though I'd be interested to hear from you if you do!)

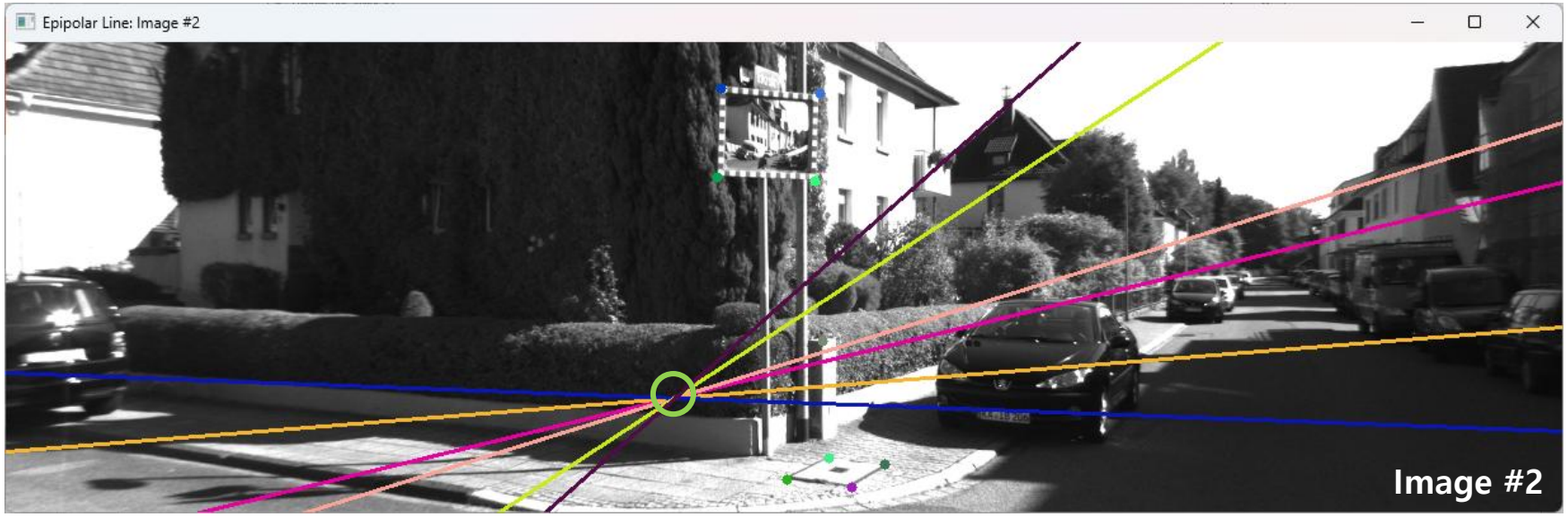
Daniel Wedge
Song: 19th October, 2008.
Video: 25th March, 2009.

danielwedge.com home

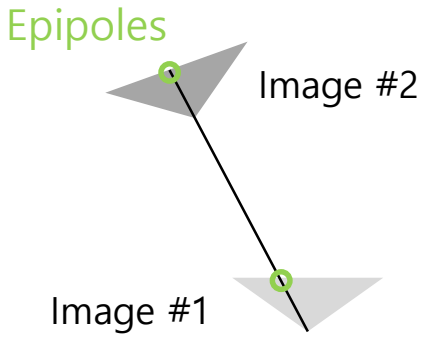
Image: Daniel Wedge

Epipolar Geometry

- Example) **Epipolar line visualization** `[epipolar_line_visualization.py]`







Epipolar Geometry

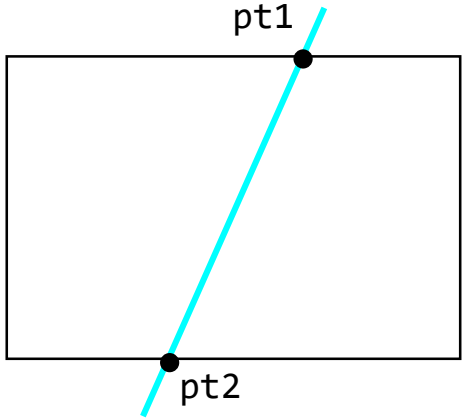
- Example) **Epipolar line visualization** [epipolar_line_visualization.py]

```
def mouse_event_handler(event, x, y, flags, param):
    if event == cv.EVENT_LBUTTONDOWN:
        param.append((x, y))

def draw_straight_line(img, line, color, thickness=1):
    h, w, *_ = img.shape
    a, b, c = line # Line: ax + by + c = 0
    if abs(a) > abs(b):
        pt1 = (int(c / -a), 0)
        pt2 = (int((b*h + c) / -a), h)
    else:
        ...
    cv.line(img, pt1, pt2, color, thickness)

if __name__ == '__main__':
    # Load two images
    img1 = cv.imread('../data/KITTI07/image_0/000000.png', cv.IMREAD_COLOR)
    img2 = cv.imread('../data/KITTI07/image_0/000023.png', cv.IMREAD_COLOR)
    # Note) `F` is derived from `fundamental_mat_estimation.py`.
    F = np.array([[ 3.34638533e-07,  7.58547151e-06, -2.04147752e-03], ...])

    # Register event handlers and show images
    wnd1_name, wnd2_name = 'Epipolar Line: Image #1', 'Epipolar Line: Image #2'
    img1_pts, img2_pts = [], []
    cv.namedWindow(wnd1_name)
    cv.setMouseCallback(wnd1_name, mouse_event_handler, img1_pts)
```



Epipolar Geometry

- Example) **Epipolar line visualization** [epipolar_line_visualization.py]

```
if __name__ == '__main__':
    # Load two images
    img1 = cv.imread('../data/KITTI07/image_0/000000.png', cv.IMREAD_COLOR)
    img2 = cv.imread('../data/KITTI07/image_0/000023.png', cv.IMREAD_COLOR)
    F = np.array([[ 3.34638533e-07,  7.58547151e-06, -2.04147752e-03], ...])

    # Register event handlers and show images
    ...

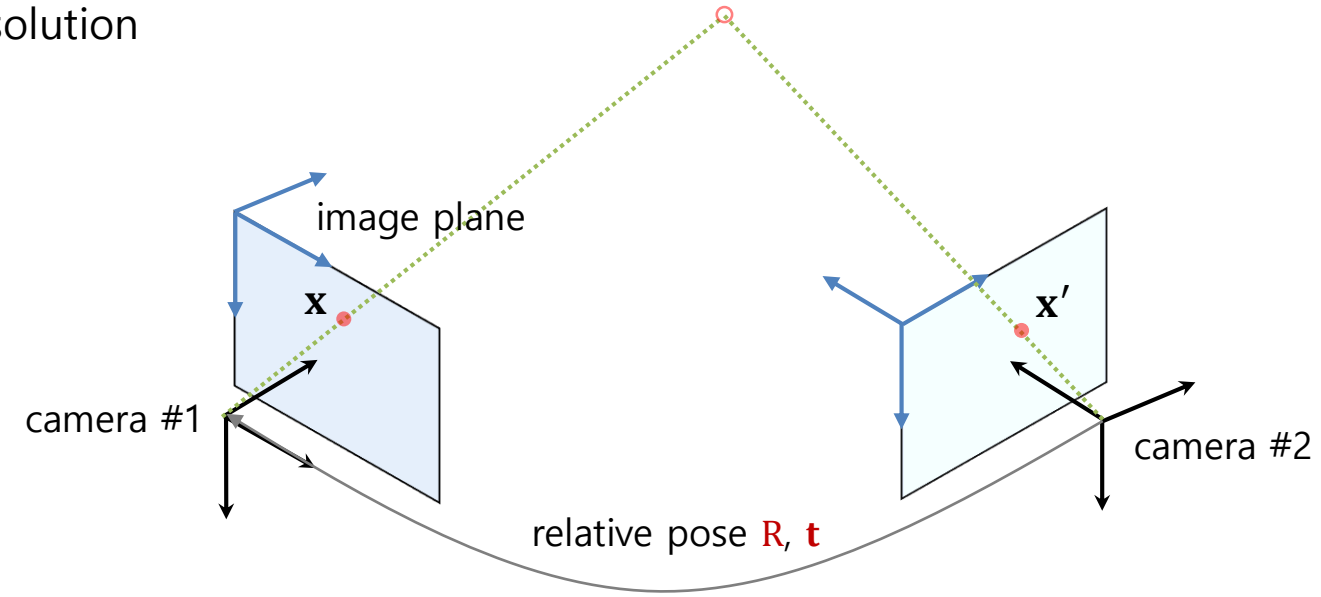
    # Get a point from a image and draw its corresponding epipolar line on the other image
    while True:
        if len(img1_pts) > 0:
            for x, y in img1_pts:
                color = (random.randrange(256), random.randrange(256), random.randrange(256))
                cv.circle(img1, (x, y), 4, color, -1)
                epipolar_line = F @ [[x], [y], [1]]
                draw_straight_line(img2, epipolar_line, color, 2)
            img1_pts.clear()
        if len(img2_pts) > 0:
            for x, y in img2_pts:
                color = (random.randrange(256), random.randrange(256), random.randrange(256))
                cv.circle(img2, (x, y), 4, color, -1)
                epipolar_line = F.T @ [[x], [y], [1]]
                draw_straight_line(img1, epipolar_line, color, 2)
            img2_pts.clear()
        cv.imshow(wnd2_name, img2)
```

$\mathbf{l}' = \mathrm{F}\mathbf{x}$

$\mathbf{l} = \mathrm{F}^{\mathrm{T}}\mathbf{x}'$

Relative Camera Pose Estimation

- **Relative camera pose estimation** (~ fundamental matrix estimation)
  - Unknown: Rotation and translation R, $\mathbf{t}$ (**5 DOF**; up-to-scale **"scale ambiguity"**)
  - Given: Point correspondence $(\mathbf{x}_1, \mathbf{x}'_1)$, …, $(\mathbf{x}_n, \mathbf{x}'_n)$ and camera matrices K, K'
  - Constraints: n x epipolar constraint ($\mathbf{x}'^{\mathsf{T}}\mathrm{F}\mathbf{x} = 0$ or $\hat{\mathbf{x}}'^{\mathsf{T}}\mathrm{E}\hat{\mathbf{x}} = 0$)
  - Solutions) **Fundamental matrix**: 7/8-point algorithm (**7 DOF**; intrinsic+extrinsic)
    - **Properties**: $\det(\mathrm{F}) = 0$ (or $\mathrm{rank}(\mathrm{F}) = 2$)
    - **Estimation**: `cv.findFundamentalMat()` → 1 solution

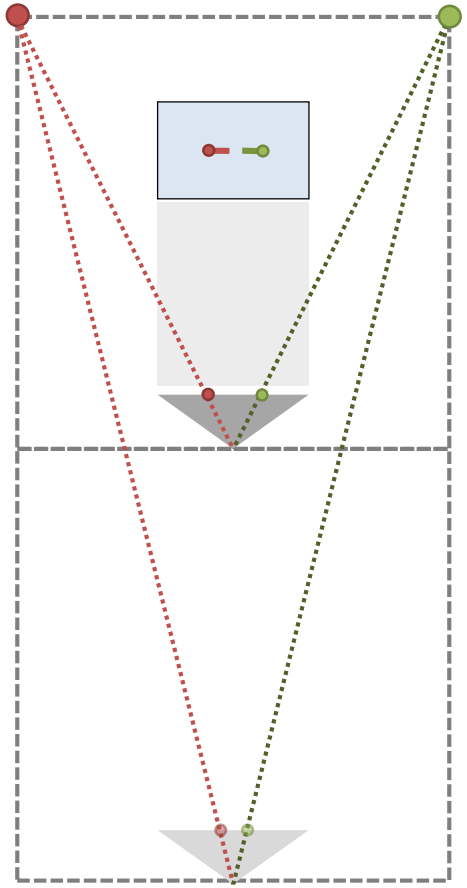


Relative Camera Pose Estimation

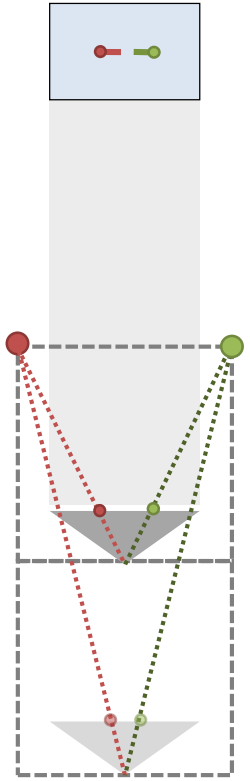
- **Relative camera pose estimation** (~ fundamental/essential matrix estimation)
  - Unknown: Rotation and translation R, $\mathbf{t}$ (**5 DOF**; up-to-scale **"scale ambiguity"**)
  - Given: Point correspondence $(\mathbf{x}_1, \mathbf{x}'_1)$, …, $(\mathbf{x}_n, \mathbf{x}'_n)$ and camera matrices K, K'
  - Constraints: n x epipolar constraint ($\mathbf{x}'^{\mathsf{T}}\mathrm{F}\mathbf{x} = 0$ or $\hat{\mathbf{x}}'^{\mathsf{T}}\mathrm{E}\hat{\mathbf{x}} = 0$)
  - Solutions) **Fundamental matrix**: 7/8-point algorithm (**7 DOF**; intrinsic+extrinsic)
    - **Properties**: $\det(\mathrm{F}) = 0$ (or $\mathrm{rank}(\mathrm{F}) = 2$)
    - **Estimation**: `cv.findFundamentalMat()` → 1 solution
    - **Conversion to E**: $\mathrm{E} = \mathrm{K}'^{\mathsf{T}}\mathrm{F}\mathrm{K}$
  - Solutions) **Essential matrix**: 5-point algorithm (**5 DOF**; extrinsic)
    - **Properties**: $\det(\mathrm{E}) = 0$ and $2\mathrm{E}\mathrm{E}^{\mathsf{T}}\mathrm{E} - \mathrm{tr}(\mathrm{E}\mathrm{E}^{\mathsf{T}})\,\mathrm{E} = 0$
    - **Estimation**: `cv.findEssentialMat()` → k solutions
    - **Decomposition**: `cv.decomposeEssentialMat()` → 4 solutions **"relative pose ambiguity"**
    - **Decomposition with positive-depth check**: `cv.recoverPose()` → 1 solution

$$\begin{matrix}\mathrm{R},\ \mathbf{t} \\ \updownarrow \quad (\mathrm{E} = [\mathbf{t}]_\times \mathrm{R}) \\ \mathrm{E} \\ \updownarrow \quad (\mathrm{E} = \mathrm{K}'^{\mathsf{T}}\mathrm{F}\mathrm{K}) \\ \mathrm{F}\end{matrix}$$

Relative Camera Pose Estimation: Scale Ambiguity



(a) 2-meter-wide tunnel



(b) 1-meter-wide tunnel



The Sandcrawler @ Star Wars IV: A New Hope (1977)

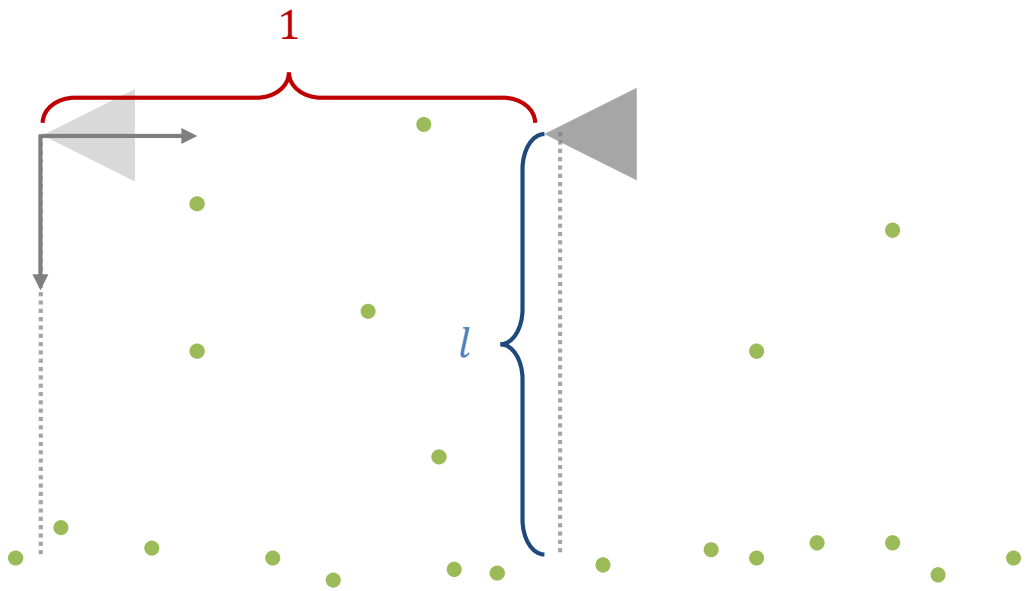




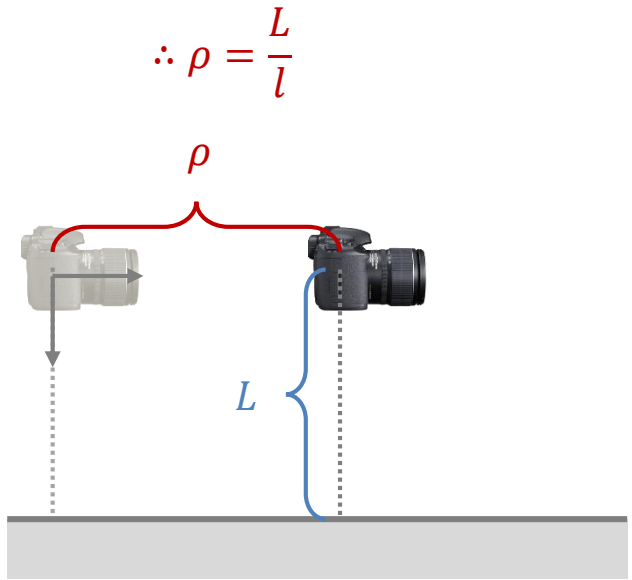
Reference: Choi et al., "Resolving Scale Ambiguity in Monocular Visual Odometry", URAI, 2013

Relative Camera Pose Estimation: Scale Ambiguity

- **How to resolve scale ambiguity**
  - **Additional sensors**: Speedometer (odometer), IMU, GPS, depth/distance (stereo, RGB-D, LiDAR, …)
  - **Motion constraints**: Known initial translation, Ackermann steering kinematics
  - **Observation constraints**: Known size of objects, known and constant height of camera



(a) The reconstructed world



(b) The real world

Reference: Choi et al., "Resolving Scale Ambiguity in Monocular Visual Odometry", URAI, 2013

Relative Camera Pose Estimation: Relative Pose Ambiguity

- **How to resolve pose ambiguity**
  - Positive depth constraint

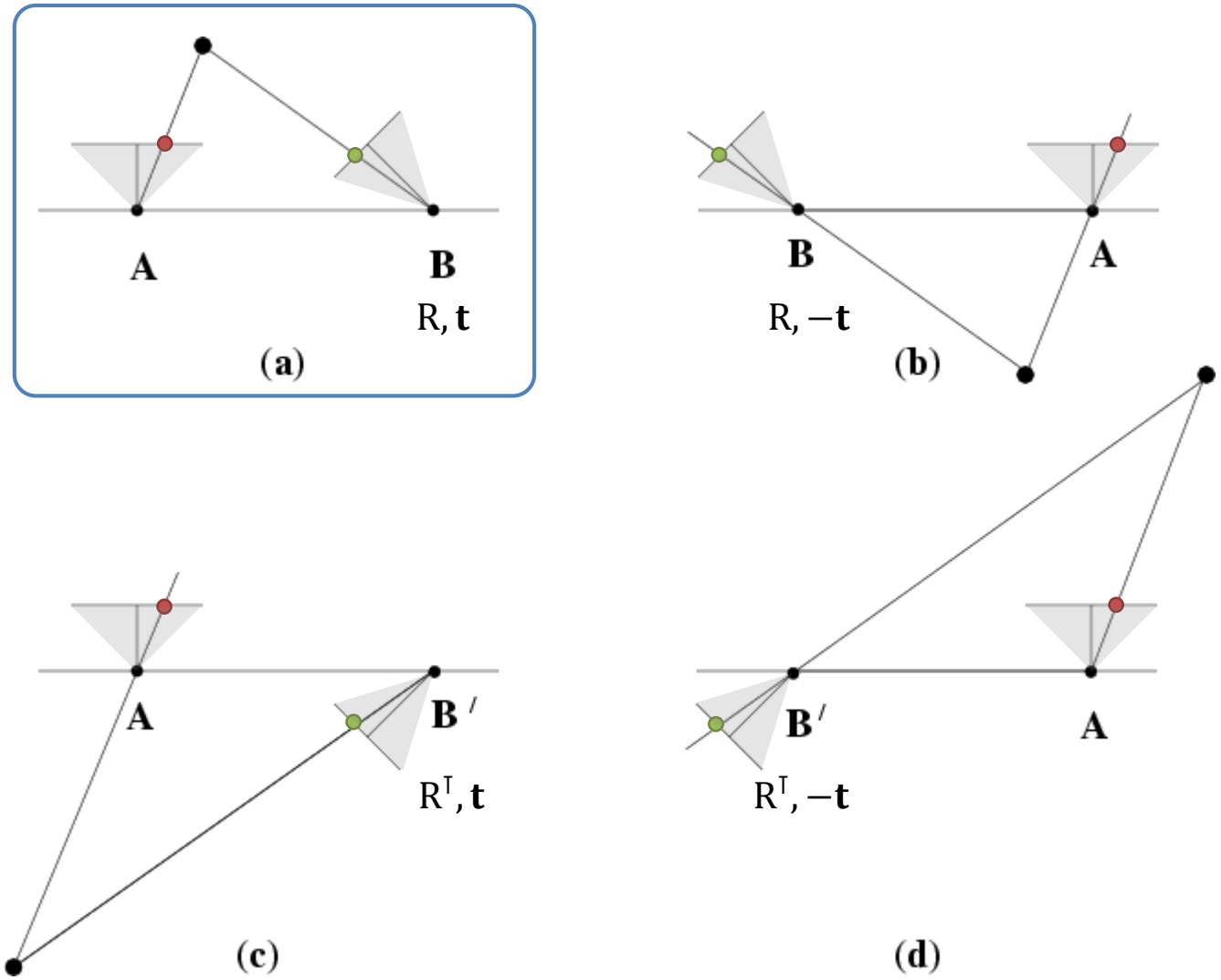


Image: Hartley and Zisserman, "Multiple View Geometry in Computer Vision" (2nd Edition), Cambridge University Press, 2004

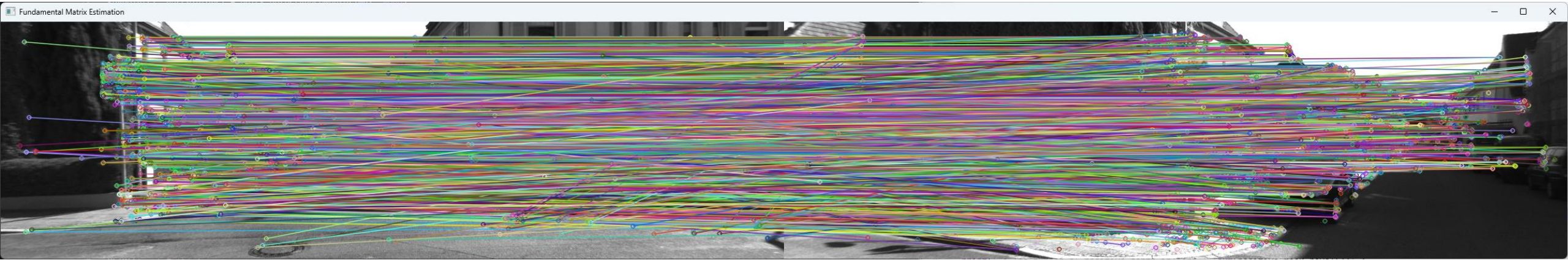
Relative Camera Pose Estimation

Example) **Fundamental matrix estimation** [fundamental_mat_estimation.py]

```
# Load two images
img1 = cv.imread('../data/KITTI07/image_0/000000.png')
img2 = cv.imread('../data/KITTI07/image_0/000023.png')
f, cx, cy = 707.0912, 601.8873, 183.1104 # From the KITTI dataset
K = np.array([[f, 0, cx], [0, f, cy], [0, 0, 1]])

# Retrieve matching points
fdetector = cv.BRISK_create()
keypoints1, descriptors1 = fdetector.detectAndCompute(img1, None)
keypoints2, descriptors2 = fdetector.detectAndCompute(img2, None)

fmatcher = cv.DescriptorMatcher_create('BruteForce-Hamming')
match = fmatcher.match(descriptors1, descriptors2)
```



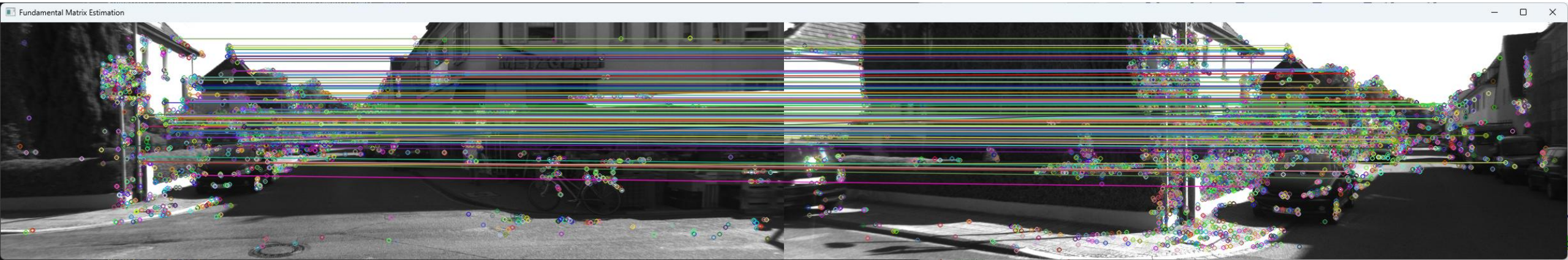
Relative Camera Pose Estimation

Example) **Fundamental matrix estimation** [fundamental_mat_estimation.py]

```
# Load two images
...

# Retrieve matching points
...

# Calculate the fundamental matrix
pts1, pts2 = [], []
for i in range(len(match)):
    pts1.append(keypoints1[match[i].queryIdx].pt)
    pts2.append(keypoints2[match[i].trainIdx].pt)
pts1 = np.array(pts1, dtype=np.float32)
pts2 = np.array(pts2, dtype=np.float32)
F, inlier_mask = cv.findFundamentalMat(pts1, pts2, cv.FM_RANSAC, 0.5, 0.999)
```



Relative Camera Pose Estimation

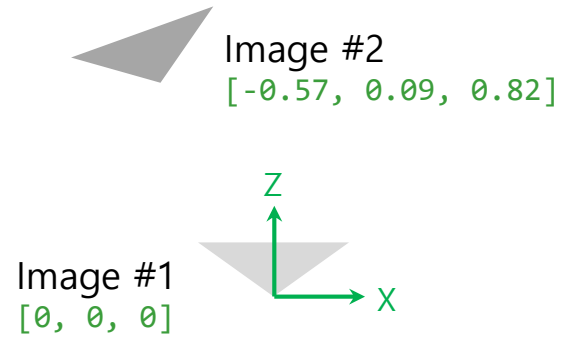
Example) **Fundamental matrix estimation** [fundamental_mat_estimation.py]

```
# Load two images
...
f, cx, cy = 707.0912, 601.8873, 183.1104 # From the KITTI dataset
K = np.array([[f, 0, cx], [0, f, cy], [0, 0, 1]])

# Retrieve matching points
...

# Calculate the fundamental matrix
...
F, inlier_mask = cv.findFundamentalMat(pts1, pts2, cv.FM_RANSAC, 0.5, 0.999)

# Extract relative camera pose between two images
E = K.T @ F @ K
positive_num, R, t, positive_mask = cv.recoverPose(E, pts1, pts2, K, mask=inlier_mask)
...
print(f'* The position of Image #2 = {-R.T @ t}') # [-0.57, 0.09, 0.82]
```



Relative Camera Pose Estimation

- Example) **Monocular visual odometry (epipolar version)** `[vo_epipolar.py]`



Given K

`Use_5pt = true`

[Tentative Matching]
- Extract optical flow between adjacent images

[Outlier Rejection]
- Remove wrong matches using `cv::findEssentialMat()`

[Relative Pose Estimation]
- Decompose camera pose using `cv::recoverPose()`

Pass tentative matches

Pass the essential matrix E

[Relative Pose Accumulation]

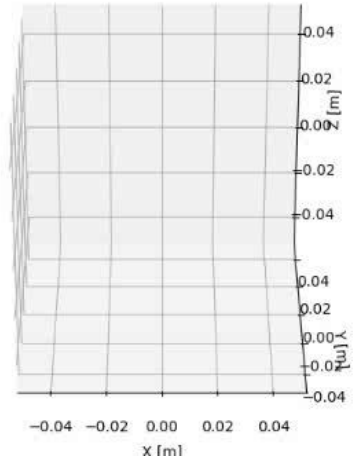
$$\mathrm{T}_n^0 = \mathrm{T}_{n-1}^0 \Delta_n^{n-1} = \mathrm{T}_{n-1}^0 \begin{bmatrix} \mathrm{R} & \mathbf{t} \\ 0 & 1 \end{bmatrix}^{-1}$$

Pass the relative pose R, **t** (Note: $\| \mathbf{t} \| = 1$)

Relative Camera Pose Estimation

- Example) **Monocular visual odometry (epipolar version)** `[vo_epipolar.py]`





(Red: Inlier, Green: Outlier)

Given K `Use_5pt = true`

[Tentative Matching]
- Extract optical flow between adjacent images

[Outlier Rejection]
- Remove wrong matches using `cv::findEssentialMat()`

[Relative Pose Estimation]
- Decompose camera pose using `cv::recoverPose()`

Pass tentative matches

Pass the essential matrix E

[Relative Pose Accumulation]

$$\mathrm{T}_n^0 = \mathrm{T}_{n-1}^0 \Delta_n^{n-1} = \mathrm{T}_{n-1}^0 \begin{bmatrix} \mathrm{R} & \mathbf{t} \\ 0 & 1 \end{bmatrix}^{-1}$$

Pass the relative pose R, $\mathbf{t}$ (Note: $\| \mathbf{t} \| = 1$)

Relative Camera Pose Estimation

- Example) **Monocular visual odometry (epipolar version)** [vo_epipolar.py]

```
video_file = '../data/KITTI07/image_0/%06d.png'
f, cx, cy = 707.0912, 601.8873, 183.1104
use_5pt = True
min_inlier_num = 100
min_inlier_ratio = 0.2
traj_file = 'vo_epipolar.xyz'

# Open a video and get an initial image
video = cv.VideoCapture(video_file)

_, gray_prev = video.read()
if gray_prev.ndim >= 3 and gray_prev.shape[2] > 1:
    gray_prev = cv.cvtColor(gray_prev, cv.COLOR_BGR2GRAY)

# Prepare a plot to visualize the camera trajectory
...

# Run the monocular visual odometry
K = np.array([[f, 0, cx], [0, f, cy], [0, 0, 1]])
camera_traj = np.zeros((1, 3))
camera_pose = np.eye(4)
while True:
    # Grab an image from the video
    valid, img = video.read()
    if not valid:
        break
```

Relative Camera Pose Estimation

- Example) **Monocular visual odometry (epipolar version)** [vo_epipolar.py]

```
# Run the monocular visual odometry
K = np.array([[f, 0, cx], [0, f, cy], [0, 0, 1]])
camera_traj = np.zeros((1, 3))
camera_pose = np.eye(4)
while True:
    # Grab an image from the video
    valid, img = video.read()
    if not valid:
        break
    if img.ndim >= 3 and img.shape[2] > 1:
        gray = cv.cvtColor(img, cv.COLOR_BGR2GRAY)
    else:
        gray = img.copy()

    # Extract optical flow
    pts_prev = cv.goodFeaturesToTrack(gray_prev, 2000, 0.01, 10)
    pts, status, error = cv.calcOpticalFlowPyrLK(gray_prev, gray, pts_prev, None)
    tracked_mask = status.ravel().astype(bool)
    pts_prev, pts = pts_prev[tracked_mask], pts[tracked_mask]
    gray_prev = gray

    # Calculate relative pose
    if use_5pt:
        E, inlier_mask = cv.findEssentialMat(pts_prev, pts, f, (cx, cy), cv.FM_RANSAC, 0.99, 1)
    else:
        F, inlier_mask = cv.findFundamentalMat(pts_prev, pts, cv.FM_RANSAC, 1, 0.99)
```

Relative Camera Pose Estimation

- Example) **Monocular visual odometry (epipolar version)** [vo_epipolar.py]

```
# Run the monocular visual odometry
while True:
    # Grab an image from the video
    ...

    # Extract optical flow
    ...

    # Calculate relative pose
    if use_5pt:
        E, inlier_mask = cv.findEssentialMat(pts_prev, pts, f, (cx, cy), cv.FM_RANSAC, 0.99, 1)
    else:
        F, inlier_mask = cv.findFundamentalMat(pts_prev, pts, cv.FM_RANSAC, 1, 0.99)
        E = K.T @ F @ K
    inlier_num, R, t, inlier_mask = cv.recoverPose(E, pts_prev, pts, focal=f, pp=(cx, cy), mask=inlier_mask)
    inlier_ratio = inlier_num / len(pts)

    # Accumulate relative pose if result is reliable
    info_color = (0, 255, 0)
    if inlier_num > min_inlier_num and inlier_ratio > min_inlier_ratio:
        T = np.eye(4)
        T[:3, :3] = R
        T[:3, 3] = t.flatten()
        camera_pose = camera_pose @ np.linalg.inv(T)
        info_color = (0, 0, 255)
```

$$\mathrm{T}_n^0 = \mathrm{T}_{n-1}^0 \Delta_n^{n-1} = \mathrm{T}_{n-1}^0 \begin{bmatrix} \mathrm{R} & \mathbf{t} \\ 0 & 1 \end{bmatrix}^{-1}$$

Relative Camera Pose Estimation

- **Relative camera pose estimation** (~ fundamental/essential matrix estimation)
  - Unknown: Rotation and translation R, $\mathbf{t}$ (**5 DOF**; up-to-scale **"scale ambiguity"**)
  - Given: Point correspondence $(\mathbf{x}_1, \mathbf{x}'_1)$, …, $(\mathbf{x}_n, \mathbf{x}'_n)$ and camera matrices K, K'
  - Constraints: n x epipolar constraint ($\mathbf{x}'^{\mathsf{T}}\mathrm{F}\mathbf{x} = 0$ or $\hat{\mathbf{x}}'^{\mathsf{T}}\mathrm{E}\hat{\mathbf{x}} = 0$)
  - Solutions) **Fundamental matrix**: 7/8-point algorithm (**7 DOF**; intrinsic+extrinsic)
    - **Properties**: $\det(\mathrm{F}) = 0$ (or $\mathrm{rank}(\mathrm{F}) = 2$)
    - **Estimation**: `cv.findFundamentalMat()` → 1 solution
    - **Conversion to E**: $\mathrm{E} = \mathrm{K}'^{\mathsf{T}}\mathrm{F}\mathrm{K}$
    - **Degenerate cases**: No translation, correspondence from a single plane
  - Solutions) **Essential matrix**: 5-point algorithm (**5 DOF**; extrinsic)
    - **Properties**: $\det(\mathrm{E}) = 0$ and $2\mathrm{E}\mathrm{E}^{\mathsf{T}}\mathrm{E} - \mathrm{tr}(\mathrm{E}\mathrm{E}^{\mathsf{T}})\,\mathrm{E} = 0$
    - **Estimation**: `cv.findEssentialMat()` → k solutions
    - **Decomposition**: `cv.decomposeEssentialMat()` → 4 solutions **"relative pose ambiguity"**
    - **Decomposition with positive-depth check**: `cv.recoverPose()` → 1 solution
    - **Degenerate cases**: No translation ($\because \mathrm{E} = [\mathbf{t}]_\times \mathrm{R}$)

R, $\mathbf{t}$
$\updownarrow$ $(\mathrm{E} = [\mathbf{t}]_\times \mathrm{R})$
E
$\updownarrow$ $(\mathrm{E} = \mathrm{K}'^{\mathsf{T}}\mathrm{F}\mathrm{K})$
F

Relative Camera Pose Estimation

- **Relative camera pose estimation** (~ fundamental/essential matrix estimation + homography estimation)

complementary

  - Solutions) **Fundamental matrix**: 7/8-point algorithm (**7 DOF**; intrinsic+extrinsic)
    - **Estimation**: `cv.findFundamentalMat()` → 1 solution
    - **Conversion to E**: $\mathrm{E} = \mathrm{K}'^{\mathrm{T}}\mathrm{F}\mathrm{K}$
    - **Degenerate cases**: No translation, correspondence from a single plane
  - Solutions) **Essential matrix**: 5-point algorithm (**5 DOF**; extrinsic)
    - **Estimation**: `cv.findEssentialMat()` → k solutions
    - **Decomposition**: `cv.decomposeEssentialMat()` → 4 solutions **"relative pose ambiguity"**
    - **Decomposition with positive-depth check**: `cv.recoverPose()` → 1 solution
    - **Degenerate cases**: No translation ($\because \mathrm{E} = [\mathbf{t}]_{\times}\mathrm{R}$)
  - Solutions) **Planar homography**: 4-point algorithm (**8 DOF**; up-to-scale **"scale ambiguity"**)
    - **Estimation**: `cv.findHomography()` → 1 solution
    - **Conversion to calibrated H**: $\hat{\mathrm{H}} = \mathrm{K}'^{-1}\mathrm{H}\mathrm{K}$
    - **Decomposition**: `cv.decomposeHomographyMat()` → 4 solutions **"relative pose ambiguity"**
    - **Degenerate cases**: Correspondence **not** from a single plane

- **Relative camera pose estimation**
  - Solutions) **Planar homography** : 4-point algorithm (**8 DOF**; up-to-scale **"scale ambiguity"**)
    - **Estimation**: `cv.findHomography()` → 1 solution
    - **Derivation**

$$\mathbf{X}' = \mathrm{R}\mathbf{X} + \mathbf{t}$$

$$\downarrow \quad \mathbf{X}' = \lambda'\hat{\mathbf{x}}' \quad \text{and} \quad \mathbf{X} = \lambda\hat{\mathbf{x}}$$

$$\lambda'\hat{\mathbf{x}}' = \lambda\mathrm{R}\hat{\mathbf{x}} + \mathbf{t}$$

$$\downarrow \quad \frac{1}{d}\mathbf{n}^{\mathsf{T}}\hat{\mathbf{x}} = 1 \quad (\because n_x\hat{x} + n_y\hat{y} + n_z\hat{z} - d = 0)$$

$$\hat{\mathbf{x}}' = \lambda''\left(\mathrm{R} + \frac{1}{d'}\mathbf{t}\,\mathbf{n}^{\mathsf{T}}\right)\hat{\mathbf{x}}$$

$$\downarrow \quad \boxed{\hat{\mathrm{H}} = \mathrm{R} + \frac{1}{d'}\mathbf{t}\,\mathbf{n}^{\mathsf{T}}}$$

$$\hat{\mathbf{x}}' = \hat{\mathrm{H}}\,\hat{\mathbf{x}}$$

$$\downarrow \quad \hat{\mathbf{x}} = \mathrm{K}^{-1}\mathbf{x} \quad \text{and} \quad \hat{\mathbf{x}}' = \mathrm{K}'^{-1}\mathbf{x}'$$

$$\mathbf{x}' = \mathrm{H}\,\mathbf{x}$$

    - **Conversion to calibrated H**: $\boxed{\hat{\mathrm{H}} = \mathrm{K}'^{-1}\mathrm{H}\mathrm{K}}$
    - **Decomposition**: `cv.decomposeHomographyMat()` → 4 solutions **"relative pose ambiguity"**
    - **Degenerate cases**: Correspondence **not** from a single plane

Relative Camera Pose Estimation: Overview

	Items	General 2D-2D Geometry	Planar 2D-2D Geometry
On Image Planes	Model	**Fundamental Matrix (7 DOF)**	**Planar Homography (8 DOF)**
	Formulation	$\mathrm{F} = \mathrm{K}'^{-\mathsf{T}}\mathrm{E}\mathrm{K}^{-1}$ $\quad \mathrm{E} = \mathrm{K}'^{\mathsf{T}}\mathrm{F}\mathrm{K}$	$\mathrm{H} = \mathrm{K}'\widehat{\mathrm{H}}\mathrm{K}^{-1}$ $\quad \widehat{\mathrm{H}} = \mathrm{K}'^{-1}\mathrm{H}\mathrm{K}$
	Estimation	- **7-point algorithm** ($n \geq 7$) → k solution - **(normalized) 8-point algorithm** → 1 solution - `cv.findFundamentalMat()`	- **4-point algorithm** ($n \geq 4$) → 1 solution - `cv::findHomography()`
	Input	- $(\mathbf{x}_i, \mathbf{x}'_i)$ [px] on the image plane	- $(\mathbf{x}_i, \mathbf{x}'_i)$ [px] on a plane in the image plane
	Degenerate Cases	- No translational motion - Correspondence from a single plane	- Correspondence not from a single plane
	Decomposition to R and $\mathbf{t}$	- Convert to an essential matrix and decompose it	- `cv.decomposeHomographyMat()`
On Normalized Image Planes	Model	**Essential Matrix (5 DOF)**	**(Calibrated) Planar Homography (8 DOF)**
	Formulation	$\mathrm{E} = [\mathbf{t}]_\times \mathrm{R}$	$\widehat{\mathrm{H}} = \mathrm{R} + \frac{1}{d}\mathbf{t}\mathbf{n}^{\mathsf{T}}$
	Estimation	- **5-point algorithm** ($n \geq 5$) → k solution - `cv.findEssentialMat()`	- **4-point algorithm** ($n \geq 4$) → 1 solution - `cv::findHomography()`
	Input	- $(\hat{\mathbf{x}}_i, \hat{\mathbf{x}}'_i)$ [m] on the normalized image plane	- $(\hat{\mathbf{x}}_i, \hat{\mathbf{x}}'_i)$ [m] on a plane in the normalized image plane
	Degenerate Cases	- No translational motion	- Correspondence not from a single plane
	Decomposition to R and $\mathbf{t}$	- `cv.decomposeEssentialMat()` - `cv.recoverPose()`	- `cv.decomposeHomographyMat()` with $\mathrm{K} = \mathrm{I}_{3\times3}$

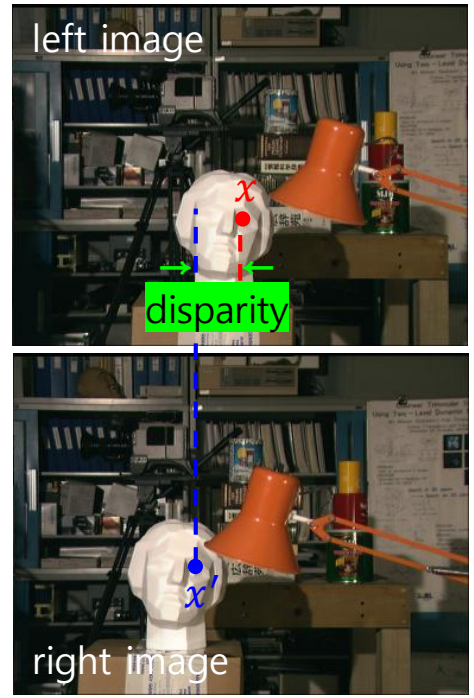
Triangulation

- **Triangulation** (point localization)
  - Unknown: Position of a 3D point $\mathbf{X}$ (3 DOF)
  - Given: Point correspondence $(\mathbf{x}, \mathbf{x}')$, camera matrices $(\mathrm{K}, \mathrm{K}')$, and relative pose $(\mathrm{R}, \mathbf{t})$
  - Constraints: $\mathbf{x} = \mathrm{K}\,[\,\mathrm{I} \mid \mathbf{0}\,]\,\mathbf{X} = \mathrm{P}\,\mathbf{X}$, $\mathbf{x}' = \mathrm{K}'\,[\,\mathrm{R} \mid \mathbf{t}\,]\,\mathbf{X} = \mathrm{P}'\,\mathbf{X}$
  - Solution
    - OpenCV `cv.triangulatePoints()`
  - Special case) Stereo cameras

$$\mathrm{R} = \mathrm{I}_{3\times 3},\ \mathbf{t} = \begin{bmatrix} -b \\ 0 \\ 0 \end{bmatrix},\ \text{and}\ \mathrm{K} = \mathrm{K}' = \begin{bmatrix} f & 0 & c_x \\ 0 & f & c_y \\ 0 & 0 & 1 \end{bmatrix}$$

where b is the baseline (distance between two cameras)

$$\therefore Z = \frac{f}{x - x'} b,\ X = \frac{x - c_x}{f} Z,\ \text{and}\ Y = \frac{y - c_y}{f} Z$$



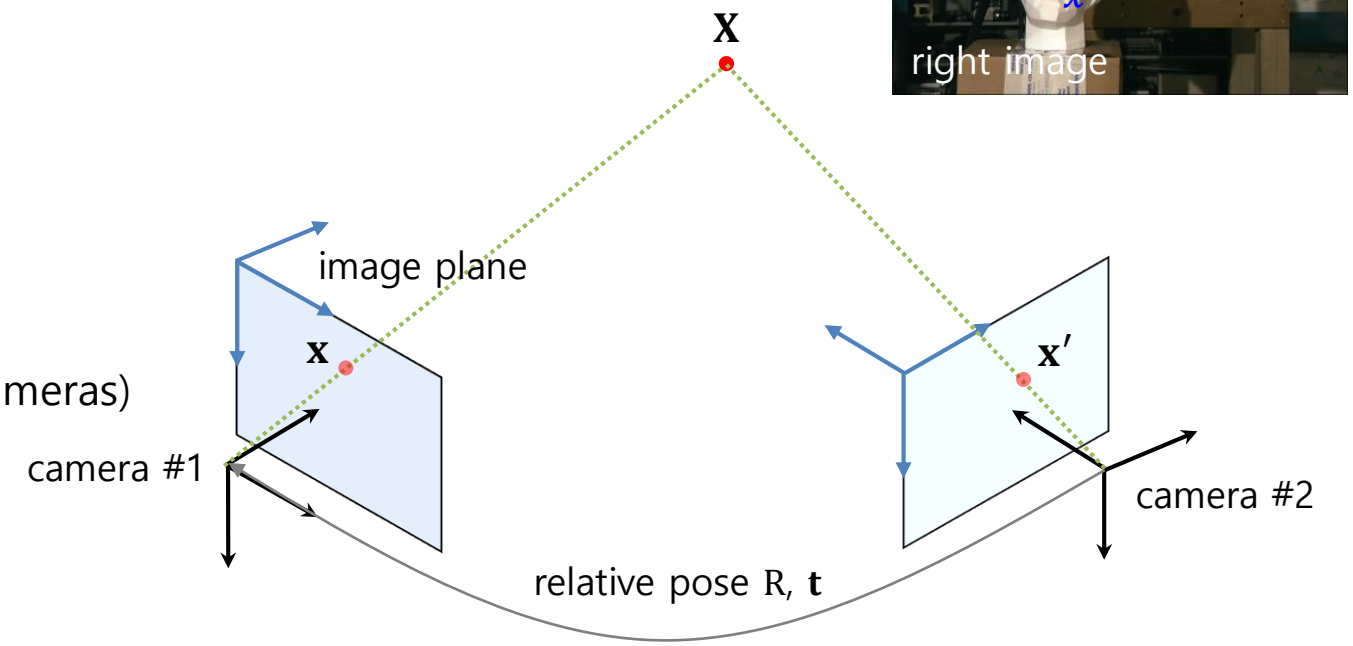


Image: Middlebury Stereo Vision Page

Triangulation

- Example) **Triangulation**

```
f, cx, cy = 1000., 320., 240.
pts0 = np.loadtxt('../data/image_formation0.xyz')[:,:2]
pts1 = np.loadtxt('../data/image_formation1.xyz')[:,:2]
output_file = 'triangulation.xyz'

# Estimate the relative pose of two views
F, _ = cv.findFundamentalMat(pts0, pts1, cv.FM_8POINT)
K = np.array([[f, 0, cx], [0, f, cy], [0, 0, 1]])
E = K.T @ F @ K
_, R, t, _ = cv.recoverPose(E, pts0, pts1, K)

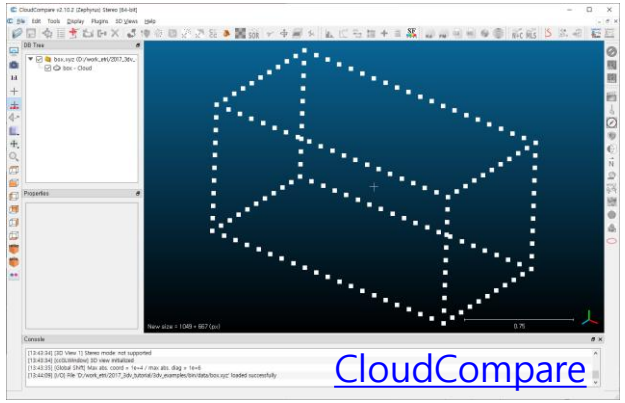
# Reconstruct 3D points (triangulation)
P0 = K @ np.eye(3, 4, dtype=np.float32)
Rt = np.hstack((R, t))
P1 = K @ Rt
X = cv.triangulatePoints(P0, P1, pts0.T, pts1.T)
X /= X[3]
X = X.T

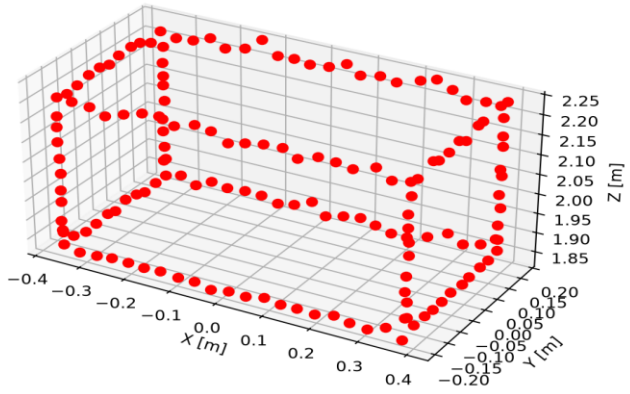
# Write the reconstructed 3D points
np.savetxt(output_file, X[:, :3])
```

$$\mathbf{x} = \mathbf{K}\,[\,\mathbf{I}\,|\,\mathbf{0}\,]\,\mathbf{X} = \mathbf{P}\,\mathbf{X}$$

$$\mathbf{x}' = \mathbf{K}'\,[\,\mathbf{R}\,|\,\mathbf{t}\,]\,\mathbf{X} = \mathbf{P}'\,\mathbf{X}$$

A point cloud: `data/box.xyz`





`output_file`: `triangulation.xyz`

Summary) Two-view Geometry

- **Planar Homography**: $\mathbf{x}_i' = \mathrm{H}\mathbf{x}_i$
  - Example) Perspective distortion correction
  - Example) Planar image stitching
  - Example) 2D video stabilization
- **Epipolar Geometry**: $\mathbf{x}'^{\mathsf{T}}\mathrm{F}\mathbf{x} = 0$ (on the image plane), $\hat{\mathbf{x}}'^{\mathsf{T}}\mathrm{E}\hat{\mathbf{x}} = 0$ (on the *normalized* image plane)
  - Example) Epipolar line visualization
- **Relative Camera Pose Estimation**: Finding R and $\mathbf{t}$ (5 DOF)
  - Solutions) **Fundamental matrix**: 7/8-point algorithm (7 DOF)
  - Solutions) **Essential matrix**: 5-point algorithm (5 DOF)
  - Solutions) **Planar homography**: 4-point algorithm (8 DOF)
  - Example) Fundamental matrix estimation
  - Example) Monocular Visual Odometry (Epipolar Version)
- **Triangulation**: Finding $\mathbf{X}$ (3 DOF)
  - Example) Triangulation

$$\begin{matrix}\mathrm{R},\ \mathbf{t} \\ \updownarrow \quad (\mathrm{E} = [\mathbf{t}]_\times \mathrm{R}) \\ \mathrm{E} \\ \updownarrow \quad (\mathrm{E} = \mathrm{K}'^{\mathsf{T}}\mathrm{F}\mathrm{K}) \\ \mathrm{F}\end{matrix}$$

Beyond Two-view Triangulation: Depth Anything V2 (2024)

- *Relative* (and also metric) **monocular depth estimation (MDE)** with DINOv2 encoder and DPT decoder via teacher/student training on heterogeneous data
  - → more fine-grained and robust than Depth Anything V1 and SD-based models (e.g. Marigold, Geowizard)
  - → more efficient (10x faster) and more lightweight than SD-based models

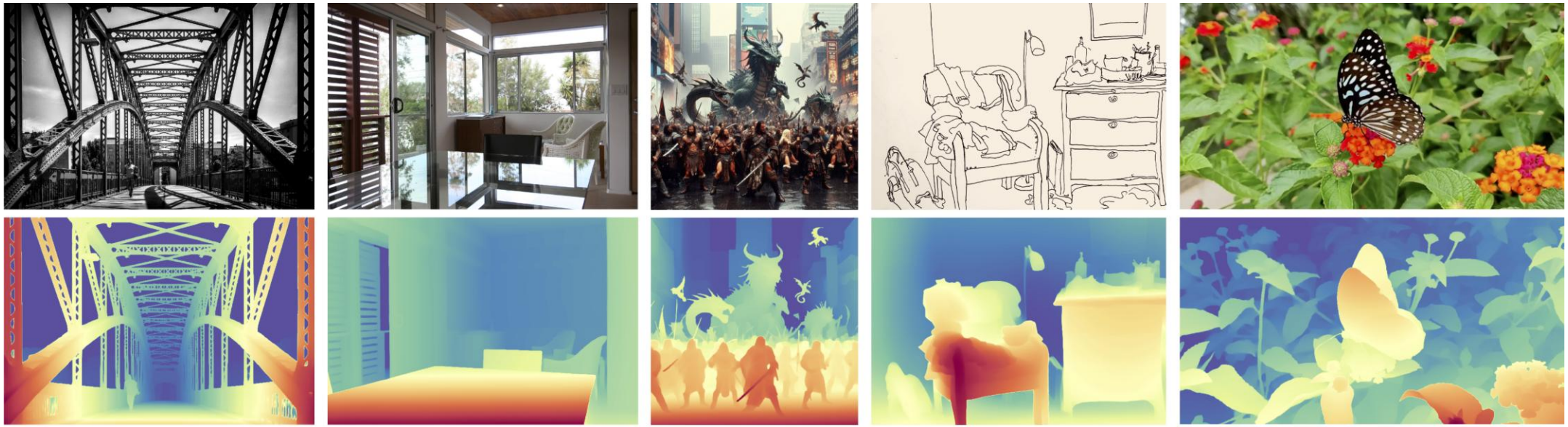


Image: Yang et al, "Depth Anything V2", NeurIPS, 2024 Homepage

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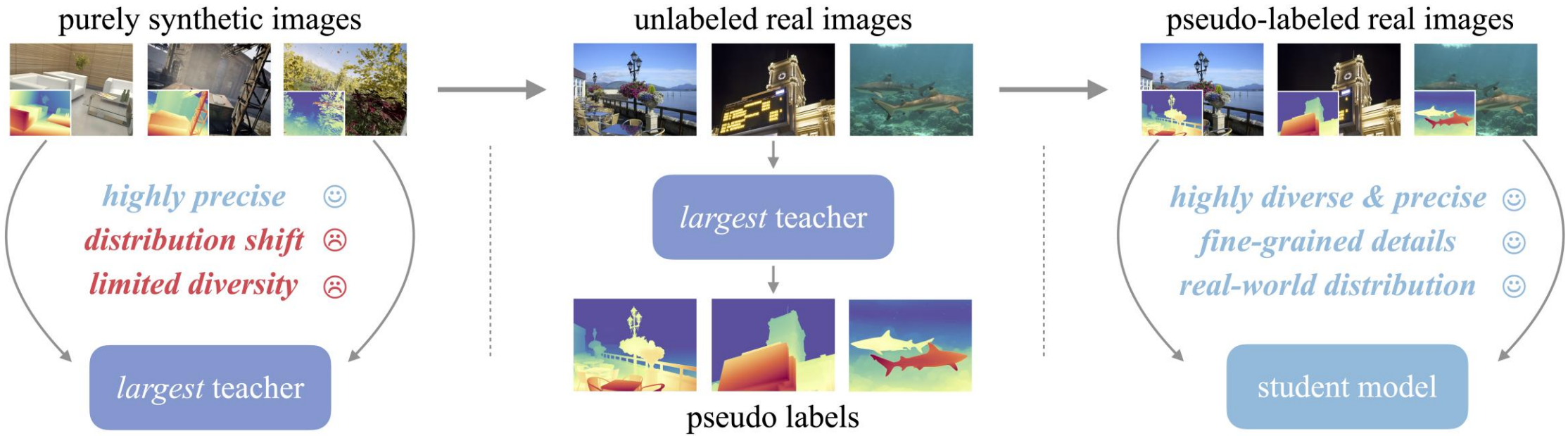


Image: Yang et al, "Depth Anything V2", NeurIPS, 2024 Homepage

Toward 3D Vision Foundation Models: DUSt3R (2024)

- *Relative* (not absolute) **two-view point maps generation** with the CroCo encoder and decoder

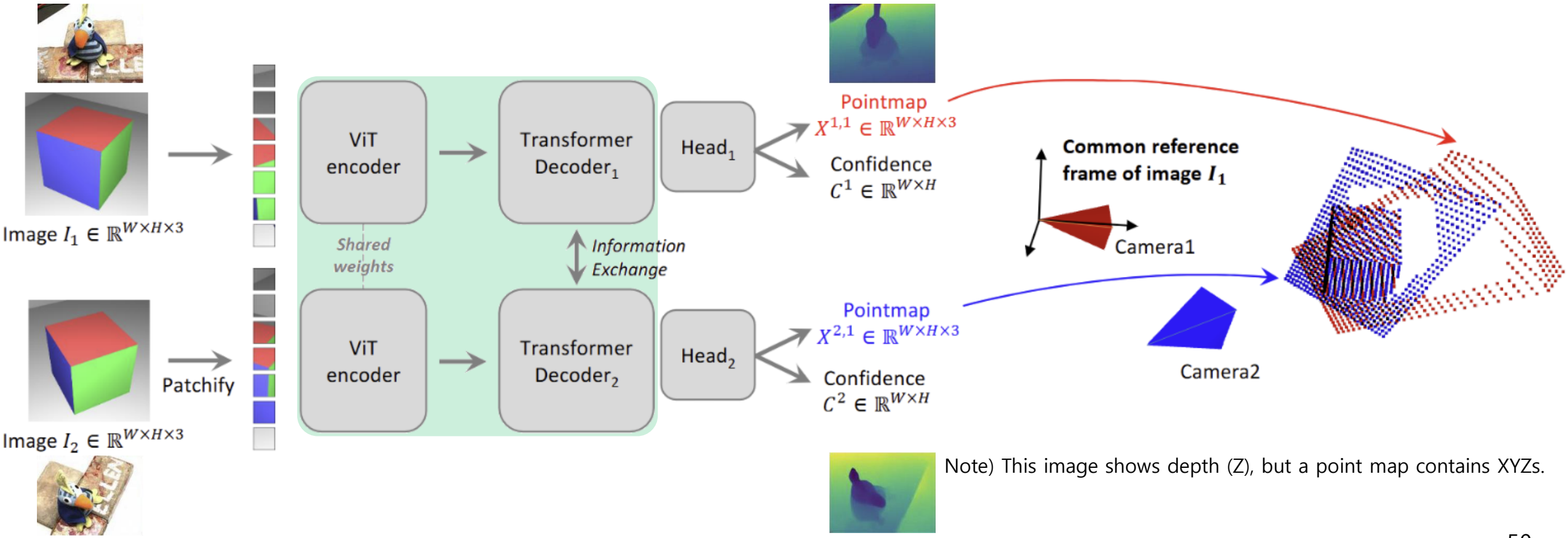
Downstream tasks

→ Point matching

→ Camera calibration, camera pose estimation (relative, multi-view, visual localization)

→ Depth estimation (monocular, multi-view), dense 3D reconstruction

Note) Scene coordinate regression (SCR): *Absolute* point map generation



Note) This image shows depth (Z), but a point map contains XYZs.

Image: Wang et al, "DUSt3R: Geometric 3D Vision Made Easy", CVPR, 2024 Homepage